00:00
This is what we're going to do today. We're going to as we also cover a couple of questions from last week, right? So the answer queries.
这就是我们今天要做的。我们还会讲到上周的几个问题，对吧？所以答案是疑问的。

00:08
Through the look there. I love you with the text analysis with Chris and.
从那边看过去。我喜欢你和克里斯的文本分析。

00:12
Thank you. And I think everyone can find that really interesting, right? Working with some of the data, looking for some of the methods and the tools. It's a useful platform, definitely if you've never done any of the text analysis work.
谢谢你！我想每个人都会觉得这很有趣，对吧？处理一些数据，寻找一些方法和工具。如果你从未做过任何文本分析工作，这绝对是一个有用的平台。

00:26
So I can highly recommend that you keep playing with it, especially if you're interested in doing text analysis. But this bit, what we're going to do is we're going to look at Special History and GS, right? We're going to work with QGS, which is the software. I hope you guys are treasure solving it on your computer.
所以我强烈建议你们继续使用它，特别是如果你们对文本分析感兴趣的话。但这一部分，我们要做的是我们要看看特殊历史和GS，对吧？我们将使用QGS，也就是软件。我希望你们珍惜在电脑上解决这个问题。

00:41
Because you'll be using activities for the exercises. The other thing you need to do for later is to download the data files. There are two ZIP folders.
因为你们会在练习中用到活动。稍后需要做的另一件事是下载数据文件。有两个ZIP文件夹。

00:52

In the first time, we cannot look at it. Please do that. We'll be using both of them today. Okay, all right. So we'll start with a little bit of a discussion about the concepts.
在第一次，我们不能看它。请这样做。我们今天会用到这两种方法。好吧，好吧。我们先来讨论一下这些概念。

01:04
Of special History in JS, right? Just to get a bearing on what these things are, right? What this data means, right? Because that's actually the biggest question that we have as historians, right? What does the data represent?
JS的特殊历史，对吧？只是为了了解这些东西是什么，对吧？这些数据意味着什么？因为这实际上是我们作为历史学家面临的最大问题，对吧？这些数据代表什么？

01:15
And GS is a very specific kind of technology, is a very specific kind of epistemology and origin. And so we've been talking about how we need to know, actually, how data is created.
GS是一种非常具体的技术，是一种非常具体的认识论和起源。所以我们一直在讨论我们需要知道，数据是如何产生的。

01:26
How data is curated, right for us to ask historical questions about it, right? This is the same as doing a provenance research for your traditional sources.
数据是如何被整理的，这样我们就可以问一些关于它的历史问题，对吧？这与为你的传统资源做来源研究是一样的。

01:36
So we'll have a look at what GS is and how we can work historically with this data, and how you can work.
所以我们会看一下什么是GS，我们如何利用这些数据进行历史研究，以及如何开展工作。

01:45

With historical sources like maps, right in a digital comment, so that.
有历史资料，比如地图，就在数字评论里，所以。

01:51
So for today. Okay.
今天就到这里。好吧。

01:54
Right. So we had a couple of questions from last week. This long one. The question is, during the collection phase of data, we often anticipate what kind of core or essential sources We need.
正确的。上周我们有几个问题。这个长。问题是，在数据收集阶段，我们通常会预测我们需要什么样的核心或基本来源。

02:06
But if they turn out to be insufficient, we may need to supplement them with less directly relevant sources.
但是，如果它们被证明是不够的，我们可能需要用不那么直接相关的来源来补充它们。

02:13
In such cases, how do we balance a proportion between the two types of materials? If the latter so less relevant of numbers the former does this affect the conclusions or accuracy of the research.
在这种情况下，我们如何平衡两种材料之间的比例？如果后者与数字的相关性较低，那么前者是否会影响研究的结论或准确性。

02:26
And it's a tough question, right? It's a question that immediately you think about when you're working with corpuses, right? Because there's so much information, there's so much of it out there.
这是个很难回答的问题，对吧？当你研究语料库时，你会立刻想到这个问题，对吧？因为有太多的信息，有太多的信息。

02:36

Some of them could be partially relevant. Some of them are directly relevant. You may find that none of them fit nicely, right, and generally for historical work. All right.
其中一些可能部分相关。其中一些是直接相关的。你可能会发现它们都不太适合历史工作。好吧。

02:47
You just include what you need in the argument, right? You don't, you might consult, say a hundred sources, right? 70. You don't say anything, so you don't really mention them at All.
你只需要在论证中包含你需要的，对吧？你不会，你可能会咨询，比如说100个信息源，对吧？70. 你什么都没说，所以你根本没提过他们。

02:57
Right, other than maybe I've looked at them and they don't say anything about what I want, right?
好吧，除了可能我看过它们它们并没有说我想要什么，对吧?

03:02
And that's kind of important, because when other historians are reading your work, they'll ask, oh, why didn't you look at this, right? So you kind of need to preempt that by saying I did, and there was nothing in it.
这一点很重要，因为当其他历史学家读到你的作品时，他们会问，哦，你为什么不看看这个？所以你需要先发制人，说我做了，里面什么都没有。

03:15
Right? And so really is thinking about how do you scope properly, and the scope really comes to your research question.
对吧?所以真正要考虑的是如何正确地界定范围，而范围与你的研究问题息息相关。

03:23
Right? What is relevant and what's irrelevant comes from your research question. So at the end of the day is asking, how much do you need to make your argument to make a defensible statement, a hypothesis.

对吧?什么是相关的，什么是不相关的，这取决于你的研究问题。所以最后的问题是，你需要多少论据才能形成一个站得住脚的陈述，一个假设。

03:35
All right, for your research, question just can't it's a little bit of intuitive thing. We learned this, and we picked this up by reading what other people have written.
好了，对于你们的研究，问题是不能的这是有点直观的东西。我们学会了这个，我们通过阅读别人写的东西来学习这个。

03:45
Right? So if you're looking for, I think this was a, this is kind of along the same lines. I'm still unsure how we can go from I have a result in, say, digital history.
对吧?所以如果你在寻找，我想这是a，这是沿着相同的路线。我仍然不确定我们如何从我有一个结果，比如说，数字历史。

03:58
So this is a historical argument, right? And this is what we're talking about in seminar two, right? How do digital insurance move from just tools?
所以这是一个历史论点，对吧？这就是我们在研讨会二中讨论的，对吧？数字保险如何从工具转变为工具?

04:07
To actual historical conclusions, right? And some of the ways of thinking about it is to consider, kind of, what this result is, right?
实际的历史结论，对吧？考虑它的一些方法是考虑，这个结果是什么，对吧?

04:18
Is it a number? Sometimes it's a number, right? It's kind of a mathematical measure of how things are related, how closely they're related, right? Or how often does something appear? So that's you.
是数字吗？有时是一个数字，对吧？它是一种数学方法，用来衡量事物之间的联系，它们之间的联系有多紧密，对吧？或者某事物出现的频率？这就是

你。

04:28
A number or some sort of fraction, right? And asking yourself, actually, what does the result mean, or does it tell us about either the data set itself or the data?
一个数字或者某种分数，对吧？问问你自己，结果是什么意思，或者它告诉我们数据集本身还是数据本身？

04:38
Itself, right? What does the trend tell you? Right? Is this a trend that points out something that we didn't really know?
本身,对吧?这种趋势告诉你什么？对吧?这种趋势会指出一些我们不知道的东西吗？

04:49
Right? Is it something that's surprising? Is it something that runs against the green of what other historians have been saying? Did you discover something you?
对吧?这是令人惊讶的事情吗？这是否与其他历史学家的观点背道而驰？你发现什么了吗？

04:58
So in digital history, we kind of do think a little bit like a scientist. I mean, historians do think like scientists, right? We do hypothesis testing and stuff. You just said now you're working with.
所以在数字历史上，我们确实有点像科学家那样思考。我是说，历史学家确实像科学家一样思考，对吧？我们做假设检验之类的。你刚说现在和你一起工作。

05:08
Different methods, and you're working at different scale, and that means you have conclusions that are based on different analysis tools.
不同的方法，你在不同的范围内工作，这意味着你的结论是基于不同的分析工具。

05:16
Right? So sometimes you do have to explain how the analysis works, and you have to explain, oh, I got one point, this 1.5 for this results, right? And you do need to explain to other historians, you don't do religion.
对吧?所以有时候你必须解释分析是如何工作的，你必须解释，哦，我得到了一分，这个结果是1.5，对吧？你需要向其他历史学家解释，你不研究宗教。

05:28
So matters what this 1.5 actually represents, right? So you might say 1.5 means actually very closely related. Most of the others are something like 0.01, so 1.5 that actually really.
所以1.5代表什么很重要，对吧？所以你可能会说1.5意味着非常接近。其他大多数都是0.01左右，所以实际上是1.5。

05:38
Debate, right? It's really significant, right? So you kind of need to do the extra translational step, but it is possible for it to fit into your historical argument.
辩论,对吧?这很重要，对吧？所以你需要做额外的平移步骤，但是它可能适合你的历史论证。

05:48
And to be a very big part of it, you just need very clear about what it actually represents and what it means for your analysis and your art conclusion.
要成为其中很大的一部分，你只需要非常清楚它实际上代表了什么以及它对你的分析和艺术结论意味着什么。

05:58
Okay. And that's this part here. What does it mean for what you're researching?
好吧。就是这一部分。这对你的研究有什么意义?

06:02

If you do any sort of computing yourself sometimes, or mathematical analysis, the computer does all the work, takes a while, pops up a little answer like 42, and then you're sitting there wondering, what does this mean?
如果你有时自己做任何计算，或者数学分析，电脑会做所有的工作，花点时间，弹出一个小答案，比如42，然后你坐在那里想，这是什么意思?

06:16
Right? Which happens all the time, right? And that's because it's been de contextualized. You're kind of running an algorithm, or you're running a pro.
对吧?这种情况经常发生，对吧？那是因为它被去语境化了。你是在运行一个算法，或者你是在运行一个专家。

06:25
A script, right? And you're generating a result, and you do need to sit and think, what does it actually say? What does it represent?
一个脚本，对吧？当你产生一个结果时，你需要坐下来思考，它到底在说什么？它代表什么?

06:34
Right, and representation of.
对，代表。

06:38
Reality, in this case, right? A number that represents something about this reality. Of the data that you're looking at is kind of something you can get used to eventually.
现实，在这种情况下，对吧？一个代表现实的数字。你所看到的数据是你最终会习惯的。

06:49
So it might look a bit boring, it might look a bit difficult at the moment, but with practice, you'll get better, and you'll be very literate.
所以它可能看起来有点无聊，可能现在看起来有点难，但通过练习，你会变

得更好，你会很有文化。

06:59
Type of results, as well as you get used to working with discounts.
类型的结果，以及你习惯使用折扣。

07:04
I foresee this will continue to kind of be on the top of your mind as you're working with different analysis and you're coming up with different results that don't look very Historical.
我预见这将继续成为你们脑海中最重要的问题，因为你们正在进行不同的分析，你们会得出不同的结果，这些结果看起来不太具有历史意义。

07:14
You always have to think, how did I get this result? And what was the matter behind it, right? What computer were we looking at? And then, therefore, what's the significance of this result.
你总是会想，我是怎么得到这个结果的？这背后有什么问题，对吧？我们在看的是什么电脑？那么，这个结果的意义是什么呢?

07:25
Okay.
好吧。

07:26
All right, any questions before we begin? Yes, do you have a question When you're creating, like, a purpose? Is there a rule of thumb against, like, combining articles from different?
好了，开始前有问题吗？是的，你有什么问题吗当你在创造目标的时候？有没有什么经验法则可以反对，比如，把不同的文章组合在一起?

07:37
News outlets, because they might have, like, slightly different kind of perspectives and things like that. Oh.
新闻媒体，因为他们可能会有稍微不同的观点之类的东西。哦。

07:43
So you have.
所以你有。

07:45
To make your own distribution, right? What might be considered a uniform corpus, right? So if you're just looking for all of nineteenth century.
做你自己的分销，对吧？什么可以被认为是统一的语料库？所以如果你只是在寻找整个19世纪。

07:55
Then you don't need a resolution of it. Enhance it. Enhances it intent. Is it important?
那你就不需要解决它了。提高它。增强它的意图。重要吗?

08:01
Right? What is your research hypothesis? Is it that?
对吧?你的研究假设是什么？是吗?

08:06
Everything in this period you can love together. Is that fine? Or are we looking at newspapers from a specific region? So maybe you're comparing with one of your neighbors. Of the analysis is.
这段时间的一切你都可以一起爱。这样行吗？还是我们在看某个特定地区的报纸？也许你在和你的邻居比较。的分析是。

08:19
Some these Asian newspapers versus American newspapers, then you don't want the London to get anything in office, right?
一些亚洲的报纸和美国的报纸，那么你就不希望伦敦得到任何东西，对吧?

08:24
So the answer always goes back to what's your research question? Is it important that we distinguish between separate sources and making

separate cultures?
所以答案总是回到你的研究问题是什么？我们区分不同的来源和创造不同的文化很重要吗?

08:34
Always go back to your question.
总是回到你的问题上。

08:38
Okay, but that's good intuition, because sometimes you want to do something at scale, right? You have to ask yourself, where is this information you're from? What data sets might be useful?
这是很好的直觉，因为有时候你想做一些大规模的事情，对吧？你得问问自己，这些信息是从哪里来的？哪些数据集可能有用?

08:49
And what can we combine, right? And this why documentation is really important. Because once you've created data set, people ask, we so why did you put these things together? And now that you put these things together, can I use it for my own project?
我们可以结合什么，对吧？这就是文档非常重要的原因。因为一旦你创建了数据集，人们会问，我们为什么要把这些东西放在一起？现在你把这些东西放在一起了，我能把它用于我自己的项目吗?

09:02
Right? Which is a question you might be asking when you find it, assets in a lot. Can I use this? Or do I need to clean it? Do I need to sort of break it apart, or you.
对吧?当你找到它的时候，你可能会问这个问题，资产很多。我可以用这个吗？还是需要我清洗一下？是我来拆散它，还是你。

09:13
Very good part of the way, so it works for my project. Okay, any other questions before we proceed.
很好的一部分，所以它适用于我的项目。好了，在我们继续之前还有问题

吗。

09:22
You. Good. Okay.
你。好。好吧。

09:24
Cool. So reminder, next week, week six is when we are doing presentations, right? These are all work, in, progress presentations. But what this means is.
酷。提醒一下，下周，第六周是我们做报告的时候，对吧？这些都是工作进展报告。但这意味着。

09:33
What you ought to be talking about. Four things, just very simple. Four things to share in your groups.
你应该说的是。四件事，很简单。在你的小组中分享四件事。

09:41
First, what are you trying to achieve, right? What is your research hypothesis or what is your public good objective, right? What are you trying to do with your project?
首先，你想要达到什么目标？你的研究假设是什么或者你的公益目标是什么？你打算怎么处理你的项目？

09:49
All right, I want to test. I want to see how, what are the perceptions of Matrasaus is critical.
好吧，我想测试一下。我想知道，对马特拉索斯的看法是如何至关重要的。

09:58
This very specific, um, that's the research hypothesis, right? Or I want to.
这个非常具体的，嗯，这是研究假设，对吧？或者我想要。

10:05

Present to the public this very complicated, fair right. I want to explain it in a way that will advance some, some public, with effective.
向公众呈现这个非常复杂、公平的权利。我想用一种能促进一些，一些公众，有效的方式来解释它。

10:14
Okay, so what's your objective? The second one is what tools and data you're working with, right? So very similar to what I ask you in week three to give me, right? I'm going to work with this corpus of newspapers, right? And I'm going to do.
好吧，那你的目标是什么？第二个是你使用的工具和数据，对吧？这和我在第三周要求你们给我的内容很相似，对吧？我将使用这些报纸语料库，对吧？我要这么做。

10:27
Sentiment analysis, right? So that's a tool that there's an approach. And my data is so a couple of newspapers from descriptive dispute, including these titles of news.
情感分析，对吧？这是一种工具，一种方法。我的数据是一些描述性争议的报纸，包括这些新闻标题。

10:39
Okay, then where are you now? Why have you gone so far? Right? So it's been three weeks since you kind of identified the project, right? What?
好吧，那你现在在哪？你为什么要走这么远？对吧?你确定这个项目已经三周了，对吧？怎么啦?

10:48
Things have you explored? What have you generally found, right? Does the data work?
你探索过什么吗？你一般都发现了什么？数据有用吗?

10:53
The tool work? Or do you realize that actually, hey, you know, I'm a bit stuck. I tried this. It's not giving me the results I'm looking for, right? And

that's fine.
工具好用吗？或者你有没有意识到，其实，嘿，你知道，我有点卡住了。我试过了。它没有给我我想要的结果，对吧？这很好。

11:03
Because that really is. What are the challenges you're facing, right? Oh, you know, the OCR is not working. The data is too dirty. I can't do any of the analysis. I realize you need to go back and do the OCR first, right?
因为这确实是。你面临的挑战是什么？你知道的，OCR坏了。数据太脏了。我不能做任何分析。我知道你得先回去做OCR，对吧?

11:13
The challenges that you're facing. Okay? You should aim for about 7 min. You can do slides if you want, as optional. We're kind of doing it because the class is split. We're going to split in two smaller groups, right? So you're going to present to your.
你所面临的挑战。好吧?你应该以7分钟为目标。你可以做幻灯片，这是可选的。我们这样做是因为这个类是分裂的。我们要分成两组，对吧？所以你要向你的。

11:26
The groups. Remember that the presentation itself is not created. The progress is not created. This is just a chance for you to get feedback from your classmates. I will group you according to the tools you're using and the data that you're using.
的组。请记住，表示本身并不是创建的。进程不是被创造出来的。这只是一个让你从同学那里得到反馈的机会。我会根据你们使用的工具和数据给你们分组。

11:38
So your classmates will be working on quite similar things. We'll be thinking about very similar issues. And the idea here is for you to throw the idea.
所以你的同学也会做类似的事情。我们会考虑非常相似的问题。这里的想法是让你抛出这个想法。

11:48
And to see how your classmates are thinking about or how they can advise you, because you'll be walking down similar parts of you. Okay.
看看你的同学是怎么想的，或者他们会给你什么建议，因为你会走到你相似的地方。好吧。

11:56
All right.
好吧。

11:58
So this is when you're doing the presentation for your project. What happens during the presentation is, while you're listening to other people's presentation, you need to do a peer review.
这是你在做项目展示的时候。在演示过程中，当你在听别人的演示时，你需要做一个同行评议。

12:10
Okay. So I will sign for you at least two other projects in your group, right? I will pair you up, essentially with other.
好吧。所以我至少会在你们组的其他两个项目上签名，对吧？我会把你和其他的配对。

12:19
Just have quite similar views as close as possible, all being about. So six, I think, right? So I've got six, and I'll tell you, okay, so.
只是有非常相似的观点，尽可能接近，都是关于。我想是6个，对吧？我有六个，我告诉你，好吧。

12:27
Visual six, you would be.
视觉6号，你是。

12:29
Anything film will be used for these two.

任何胶片都可以用来拍摄这两张照片。

12:31
And what I want you to do is I want you to pay attention to the method and the approach, right? Because these are kind of fundamental questions that determine whether project succeeds or fail.
我想让你们做的是我想让你们注意方法和途径，对吧？因为这些都是决定项目成败的基本问题。

12:42
I want you to think about what kind of practical solutions you can give based on the project, and what is check.
我想让你考虑一下你可以根据这个项目给出什么样的实际解决方案，以及什么是检查。

12:51
Ah. So a lot of corpuses are really interesting. A lot of the essays are really interesting. They can go many different ways, but you have to think about, okay, what is it that my classmate wants to achieve.
啊。所以很多语料库都很有趣。很多文章都很有趣。他们可以有很多不同的方式，但你必须考虑，好吧，我的同学想要实现什么。

13:01
Right? Put yourself in their shoes. How would you do it? Right?
对吧?设身处地为他们想想。你会怎么做？对吧？

13:04
There's a rubric that's in the syllabus, right? This is, this is the only part that's created about next week is your peer review, the presentations and cells progress not created.
大纲里有一个大纲，对吧？这是，这是下周唯一创建的部分是你们的同行评议，报告和细胞进展没有创建。

13:14
But your reviews are created. Okay.

但是你的评论是被创造出来的。好吧。

13:17
All right. Reviews, generally useful ones, right? That are actionable. They have good advice and provide examples about what people can look at, right? Or links to articles like that.
好吧。评论，通常是有用的，对吧？这些都是可操作的。他们有很好的建议，并提供了人们可以看的例子，对吗？或者链接到类似的文章。

13:28
Are generally about a page, single, spaced. Okay, so maybe came for that. I've seen reviews that call up to three pages because, you know, they're very close.
一般为一页左右，单页，行距。好吧，也许是为了这个。我曾见过长达三页的评论，因为，你知道，它们非常接近。

13:37
There are methods that they want to explain a bit more. That's fine.
有些方法他们想多解释一下。这很好。

13:42
Where below one page is generally maybe a bit too little in terms of the detail, in terms of how deeply you're interrogating, kind of the method.
一页纸以下的内容就细节而言，就你审问的深度而言，就方法而言，可能有点太少了。

13:53
So in or about any.
所以in或about any。

13:56
Okay, our class is Tuesday, so actually, I might shift is from Wednesday to next Friday instead.
我们的课是周二，所以我可能会把周三改到下周五。

14:05
All right, any questions about the peer reviews.
好了，关于同行评议有什么问题吗。

14:11
Other presentations.
其他报告。

14:14
Good.
好。

14:19
Sorry sorry.
对不起对不起。

14:26
On the Friday if you want to turn on Sunday or But that's the test week.
在周五，如果你想在周日或者那是考试周。

14:37
I mean, I don't want to touch your recess week, but maybe you should be taking a break during recess week.
我是说，我不想干涉你的休息周，但也许你应该在休息周休息一下。

14:45
Next After the after Not like not today. Like not tomorrow.
下一个以后以后不像今天。比如明天。

14:52
So next week's presentation on Tuesday, and then.
下周二的演讲，然后。

14:57
To under Friday, or then.

到星期五，或者到那时。

14:59
Because ideally you're really making so all you're doing for the assumption is filling out.
因为理想情况下你所做的一切都是为了满足假设。

15:09
Writing it out. I mean, if you need extra time, let me know, right?
把它写出来。我是说，如果你需要额外的时间，告诉我，好吗?

15:14
To do the reviews, but you ought to be doing so actively noting our suggestions and things.
做评论，但你应该积极地做，注意我们的建议和事情。

15:22
You can look at things after, which is why I don't ask you to show me at the end of that.
你可以看之后的东西，这就是为什么我不要求你在最后给我看。

15:27
I see. Oh, I wanted to find an example that I saw.
我明白了。哦，我想找一个我看到的例子。

15:34
And we should, like, have two things we need to solve.
我们应该有两件事需要解决。

15:35
Ship it up. Okay.
把它运上去。好吧。

15:37
But again, I'm very flexible in terms of the lines. If you are pushing for

something else and you have like work or other courses, whatever, let me know.
但是，我在线条方面很灵活。如果你想争取别的东西，比如工作或其他课程，无论如何，请告诉我。

15:47
But we do want to make it a bit tiny, because this will go to your company, right?
但我们确实想把它做得小一点，因为这是给你们公司的，对吧?

15:53
And that's open.
这是开放的。

15:54
So you will receive two pieces of peer review.
所以你会收到两份同行评议。

15:58
I will have a look at and I will comment be like, okay.
我会看一看，然后评论说，好的。

16:09
Okay. Any other questions about next week? We're just doing presentations since we just no readings.
好吧。关于下周还有问题吗？我们只是做陈述，因为我们没有阅读材料。

16:23
Because right now we only are exposed to two different digital tools. What if one of our classmates want to do something that we haven't even cut up on yet in our readings. How do we really evaluate energy? So I would.
因为现在我们只接触到两种不同的数字工具。如果我们的一个同学想做一些我们在阅读材料中还没有提到的事情怎么办？我们如何评估能量呢？所以我会。

16:37
Try and group everyone according to what they are working on. So if, let's say there's a group that's doing schoolly telling E.G., a group, you all together.
试着根据每个人正在做的事情对他们进行分组。假设有一群人在学校里讲故事，比如，你们都在一起。

16:48
So in some ways, you would all have some you have started looking at all the stuff before we've actually done that particular seminar.
所以在某种程度上，你们都有一些你们已经开始看所有的东西在我们真正做那个特定的研讨会之前。

16:55
That's so does it make sense? So if you're thinking about working with someone else tree, then, you know, this week.
这有意义吗？所以如果你想和别人一起工作，那么，你知道，这周。

17:05
For the group. Yes, if you haven't told me anything, we kind of changed a little bit.
为了团队。是的，如果你什么都没告诉我的话，我们有点改变了。

17:12
Did you change partners, or did you change project? We didn't grow up, okay, before, now we wanted to group up. Okay, okay. Well, let me join me an email so I Can.
你是换了合作伙伴，还是换了项目？我们以前没有长大，好吗，现在我们想组队。好吧,好吧。好吧，让我发邮件给我，这样我就可以。

17:23
Okay, my.
好的,我的。

17:24

So I'll make sure you're listening to your bottle is like accidentally.
所以我要确保你在听你的瓶子是不小心的。

17:29
All right, any other questions. So about next week.
好，还有问题吗？那么下周吧。

17:37
Oh.
哦。

17:37
So I've had to look at everyone's projects for both projects and data that you're working on. This is the order of the seminars after the research week, right?
所以我不得不查看每个人的项目，包括你正在处理的项目和数据。这是研究周之后的研讨会顺序，对吧？

17:49
So this week is spatial. Next week is plantations and a research week. And then what we look at in week seven is we are going to look at AI assisted research. How can you use?
这周的主题是空间。下周是种植园和研究周。第七周，我们将学习人工智能辅助研究。如何使用？

17:58
In this case, gen AI to help you progress with some of the work that we do. Assistments.
在这种情况下，人工智能可以帮助你完成我们所做的一些工作。Assistments。

18:03
We will look a little bit into quantitative because I know some of you are looking at statistical stuff as well.
我们会稍微研究一下定量因为我知道你们中的一些人也在研究统计方面的东

西。

18:11
So we'll have a little look at that. Week eight is when we do visualizing data. So graphic representations, generally that and relationships.
我们来看一下。第八周是我们可视化数据的时候。一般来说，图形表示和关系。

18:23
Network analysis. So we look at that on week eight. Week nine is the sandbox session. Sandbox is where you bring.
网络分析。我们在第八周看这个。第九周是沙盒环节。沙盒是你带来的地方。

18:32
Issues. And again, we will come back and travel together. So we'll probably bring into teams again, the same themes as before, and you can work together.
问题。再一次，我们会回来一起旅行。所以我们可能会再次组队，和之前一样的主题，你们可以一起工作。

18:41
To see where they can resolve each other's issues.
看看他们能在哪里解决彼此的问题。

18:44
We're all in the room together, so we are.
我们都在一个房间里，所以我们是。

18:49
All have a lot of different expertise, a lot of different experience, so it's unlikely that we won't be able to solve your problem. Okay.
他们都有很多不同的专业知识，很多不同的经验，所以我们不太可能解决不了你的问题。好吧。

18:55
So that's week nine. So some box has session. If we have time, we can talk a little bit about media preservation, because.
这是第九周。有些盒子有session。如果我们有时间，我们可以谈谈媒体保护，因为。

19:02
So that's kind of a auxiliary topic that kind of sits on the side, that goes beyond just tax itself, but things like video, video formats, audio formats, right?
所以这是一个次要的话题，不只是税收本身，还有视频，视频格式，音频格式，对吧?

19:14
Are and will be very important sources of historical information, and they are kind of separate issues there.
现在和将来都是非常重要的历史信息来源，它们是不同的问题。

19:22
Okay. And every ten we'll do school e telling, just based on what everyone's working on. But remember, if you're looking on a later session.
好吧。每隔10分钟，我们会根据每个人的学习情况，做一次学校汇报。但请记住，如果你想看以后的治疗。

19:32
So like we haven't covered it in class, feel free to look at that material first if you want to predict that primarily.
就像我们在课堂上没有讲到的那样，如果你想主要预测的话，可以先看一下这些材料。

19:42
Right. Come see me if you need help getting started on any of those concepts.
正确的。如果你需要帮助来开始学习这些概念，可以来找我。

19:47
Okay. And then week eleven and twelve, we reserve for final presentations. So I'll ask you guys who wants to present first. Some of you may want to present first, just so you have a bit more time to.
好吧。然后第11周和第12周，我们留作期末报告。我来问你们谁想先发言。你们中的一些人可能想先做陈述，这样你们就有更多的时间。

20:04
Do the critical reflection essay at the end, which is due in week. So.
最后写一篇批判性反思文章，一周内交。所以。

20:08
You go. Some of you may want to delete you. We talk to your communications.
你走。你们中的一些人可能想删除你。我们与你的通讯进行对话。

20:14
So I'll have the same option for that. All right. And then we put in, we run up and say goodbye.
所以我有相同的选项。好吧。然后我们放进去，我们跑上去说再见。

20:21
Very fast. Okay, let's get the limit of today. So what we're going to do today is we're going to look at some of the theoretical implications of spatial.
非常快。好的，我们来求一下今天的极限。所以我们今天要做的是我们要看看空间的一些理论含义。

20:33
Geographical history, right? What happens when we think about spatial history and incorporate notions of space and place right into our historical work?
地理历史，对吧？当我们思考空间历史并将空间和地点的概念纳入我们的历史工作时会发生什么？

20:43
We will look at some of the tools and the data that we use in what we call historical GS, right? So GS for historical methods, or?
我们会看一些工具和数据我们称之为历史GS，对吧？GS代表历史方法?

20:53
Historical material, put in JS formats or rationalized to GS.
历史资料，放入JS格式或合理化到GS。

20:59
And to help us with that, we're going to do two exercises. We're going to do what we call geo referencing, right?
为了帮助我们理解，我们要做两个练习。我们要做的是地理引用?

21:05
And then this girl is going to do so. Something further down the background is producing a map based on GSB.
然后这个女孩就会这么做。后台的某个东西正在生成一张基于GSB的地图。

21:11
Okay.
好吧。

21:13
And these are the questions I know many of you wrote a response for this week. How can special did I help you as a historian, right? And what considerations do you Have.
我知道你们很多人这周都为这些问题写了回复。作为历史学家，我能帮到你什么吗？你有什么考虑?

21:25
Maybe use issue leader. Um.
也许可以使用issue leader。嗯。

21:28

Anyone have any particular thoughts before we start from special data.
在我们从特殊数据开始之前有人有什么特别的想法吗?

21:34
Okay, others will start looking at some of the basics.
好的，其他人会开始看一些基础知识。

21:44
How does your hands represent reality?
你的手是如何表现现实的?

21:50
What does it do?
它是做什么的?

21:51
Giants can perform the readings. I think the first one is talk about how giant architecture movements. So it's like also studying how from point axis text both.
巨人可以进行阅读。我想第一个是关于巨型建筑是如何运动的。这就像研究如何从点轴文本。

22:03
And also different multiple points as well. Okay. So what is it, how is it structured? GS data.
也有不同的多个点。好吧。那么它是什么，它是如何构成的？GS数据。

22:12
What does guys didn't look like?
男人长得不像什么?

22:17
Sorry, good. 1.3.
对不起,很好。1.3.

22:24
Vista and Rasper.
Vista和Rasper。

22:33
Interesting. Okay, um, we'll come to that concept.
有趣。好吧，我们会讲到这个概念的。

22:40
All jazz data is.
所有jazz数据都是。

22:43
Is it true?
这是真的吗?

22:47
Yes, yes. All j data is probably back to data, but JS data.
是的,是的。所有j数据可能都回到了数据，但是JS数据。

22:56
It comes in tree palms. We've got a point.
它来自棕榈树。我们说的有道理。

23:01
You've got a line.
你排队了。

23:06
Have you got what you're saying, a polygon.
你知道你说的是什么吗，一个多边形。

23:13
So that's reality. JS Reality is three types.
这就是现实。JS现实有三种类型。

23:24
Of things, coins, lines of polygons.
事物，硬币，多边形的线条。

23:32
What does this imply?
这意味着什么?

23:36
I see.
我明白了。

23:40
But what does this mean for how we describe reality?
但这对我们如何描述现实意味着什么呢?

23:53
What happens when we reduce reality into swings and polygons?
当我们将现实简化为摆动和多边形时会发生什么?

24:00
It bites and numbers, right? Essentially, one and zero, right? Is it there? Is it not there?
它咬人又咬人，对吧？本质上是1和0，对吧？它在那里吗？它不在那里吗?

24:13
Ask your question. Is Singapore a point, a line, or polygon? Polygon?
问你的问题。新加坡是一个点、一条线还是一个多边形？多边形?

24:21
Very historian of you. Okay, so.
你真是个历史学家。好的,所以。

24:35

Actually noticed.
注意到。

24:45
What's the matter?
怎么了?

24:45
The map of the UK and the period that I study, right? How do you think it was me?
英国的地图和我研究的时期，对吧？你怎么会觉得是我干的?

24:56
Oh.
哦。

24:56
What do you think was me?
你以为是我干的?

25:00
Lines, right? You kind of put lines together, right? And then you use lines, you kind of draw out.
行,对吧?你把线条放在一起，对吧？然后用线，把它画出来。

25:10
The country, right? Use the lines to draw the boundaries and then indicate which region is which, right?
国家，对吧？用这些线来画边界，然后指出哪个区域是哪个区域，对吗?

25:20
So what happens when you draw lines?
那么当你画线的时候会发生什么呢?

25:25

Are these things accurate?
这些东西准确吗?

25:28
You're shaking it. I do not here.
你在摇晃它。我不知道。

25:34
It's not the real thing. It's not a real thing, right? This is not the country itself, right? We're trying to represent something in a symbolic form.
这不是真的。这不是真的，对吧？这不是国家本身，对吧？我们试着用符号的形式来表示一些东西。

25:50
Right? We've been talking about how, in digital history we mediate, right? When you digitize something, you are actually removing and abstracting one layer of it, right?
对吧?我们一直在讨论，在数字历史中我们是如何调解的，对吧？当你把某样东西数字化时，你实际上是在移除和抽象它的一层，对吧?

26:00
We say that the digitalized format is a representation of the original sources of right? So a map, right? This.
我们说数字化格式是原始来源的一种表现，对吗？地图，对吧？这一点。

26:09
It's not actual country, right?
不是真正的国家，对吧?

26:13
It can be useful information because, you know, it's telling us something, right? It's giving us all sorts of different pieces of information about this place, right?
它可能是有用的信息，因为它告诉了我们一些东西，对吧？它给了我们关于这个地方的各种不同的信息，对吧?

26:22
But it's not the thing in and of itself. So when I ask you a Singapore, a polygon, or a point, where do you guys choose polygon?
但这不是事情本身。所以当我问你们一个新加坡，一个多边形，或者一个点，你们会选择哪个多边形?

26:31
Why is it not? Point?
为什么不是呢？点吗?

26:34
Isn't it all? Abutary Yes, it is. How would you decide?
这不是全部吗？阿布塔利：是的。你会怎么决定?

26:42
So it depended on the scale of the map precisely right. It does it matter?
所以这取决于地图的比例是否准确。这有关系吗?

26:50
Whether, when does it matter that you need a polygon?
什么时候需要一个多边形呢?

26:55
Right? When does it matter that you need it to be a point?
对吧?什么时候你需要它成为一个点才重要呢?

27:00
So we go back to your research question. Is it important for the skill?
所以我们回到你的研究问题。它对技能很重要吗?

27:10
Is it important for your analysis, right? Like, let's say we have a.
这对你的分析很重要，对吧？比如说，我们有一个。

27:17
Let's say we have a war map. Let's see what'll be Your point will be a polygon.
假设我们有一张战争地图。我们来看看你的点是一个多边形。

27:23
Please speak.
请说话。

27:29
Right?
对吧?

27:29
And look for ugly.
寻找丑陋之处。

27:32
Right? So let's see simple. Then the US is.
对吧?我们来看看简单的。然后是美国。

27:39
Representation of reality, right? But if you're going to complete the distance.
现实的表征，对吧？但是如果你要走完全程。

27:51
Pin Season Point, new York, right? Would you use a polygon? You just use a point, right? And you draw a direct line in between.
Pin Season Point，纽约，对吧？你会使用多边形吗？你只用一个点，对吧？然后在两者之间画一条直线。

28:01
And you counted, right? It was way less complicated, right? Than using a polygon. Okay.

你数过了，对吧？其实没那么复杂，对吧？比使用多边形要好。好吧。

28:08
So the answer usually is it depends on what you're asking and what you're trying to do. Whether something is a point or line or polygon is determined by you.
所以答案通常是，这取决于你问的是什么，你想做什么。某物是点、线还是多边形由你决定。

28:18
About curating that data. Again, you did the same thing for digitization. And here is exactly the same thing.
管理这些数据。再一次，你对数字化做了同样的事情。这里是完全一样的。

28:26
The GS is just another form of mediation.
GS只是另一种形式的调解。

28:32
Here we go.
开始吧。

28:32
So when you're working in GS data said, you ask yourself, why is it that Singapore is a polygon in this one?
所以当你在处理GS数据时，你会问自己，为什么新加坡在这个图中是一个多边形?

28:40
Now, if we had Singapore, which I would just write the diamond.
现在，如果我们有新加坡，我就把它写成钻石。

28:46
Rainbow, um, right.
彩虹，嗯，对。

28:53
This might be Singapore, right?
这可能是新加坡，对吧?

28:57
Do you include the islands?
包括岛屿吗?

28:59
The southern island, santosa, strong Island. So there are some offshore islands. For those are not familiar with Singapore geography, the Strong Island here.
南部岛，圣多萨岛，强岛。这里有一些近海岛屿。对于那些不熟悉新加坡地理的人来说，这里是强岛。

29:09
Also here. That is the home here. Loneliness.
也在这里。那是我们的家。孤独。

29:14
Does the polygon includes them, is it put out here?
这个多边形包含它们吗，它放在这里了吗?

29:21
This is random island. As I see.
这是随机岛。正如我所见。

29:27
Do we include those in the polygon or Singapore?
我们把这些包括在多边形或新加坡吗?

29:31
Yes, no, tough question, right? Yes. Why? Yes.
是，不是，很难回答的问题，对吧？是。为什么?是。

29:39
Because it has job.
因为它有工作。

29:43
Because, because these are included.
因为，因为这些都包括在内。

29:50
Right?
对吧?

29:52
Right? This is part of the country, right? So then what we're looking at is we're kind of looking at the international boundaries of.
对吧?这是国家的一部分，对吧？所以我们要看的是我们要看的是国际边界。

30:03
This country, and that is.
就是这个国家。

30:06
So if you're pluxing part into Google Network again.
如果你再次加入b谷歌网络。

30:13
You get a polygon, or you get a point.
你得到一个多边形，或者一个点。

30:21
Get a map.
拿张地图。

30:21

Nowadays, you get a polygon right last time. If you were to plot in Singapore, what you get is to get somewhere in the middle of magazine.
现在，上次你得到了一个正确的多边形。如果你在新加坡策划，你得到的是在杂志中间的某个地方。

30:31
And it still happens today. For some map sentences, they don't give you polygons. They give you the center of the polygon.
时至今日，这种情况仍在发生。对于一些地图句子，它们不会给你多边形。给出了多边形的中心。

30:44
Oh.
哦。

30:44
So you take this polygon and you can actually automatically complete with the center of it.
所以你取这个多边形你可以自动完成它的中心。

30:54
Yes.
是。

30:54
I see a few confused pieces.
我看到了一些混乱的片段。

30:57
What are you confused about?
你在困惑什么?

31:06
Nine.
九个。

31:06
Okay. Points. One's polygons.
好吧。点。一个人的多边形。

31:12
Any problems with that.
有什么问题吗?

31:15
What are lines? What are lines composed of?
什么是线？线是由什么组成的?

31:20
So interestingly, interestingly, life are not necessarily affect.
所以有趣的是，生活不一定会受到影响。

31:31
Because infinity is very difficult to calculate computationally. So what lines are?
因为无穷是很难计算的。那么直线是什么呢?

31:36
A point.
一个点。

31:38
So we use another color.
所以我们用另一种颜色。

31:46
Anyone here knows their coordinate on coordinate.
每个人都知道他们的坐标。

31:57

There's a comma. There are two numbers, right? There's a comma. All right. So I've got a point here.
有一个逗号。有两个数字，对吧？有一个逗号。好吧。这里有一个点。

32:06
If I draw a graph like that, this is one one, let's say.
如果我像这样画一个图，这是1。

32:12
So if I look at my JS data, it says Point, and the coordinate is one one.
如果我看JS数据，它说点，坐标是1。

32:22
Now, if I drew.
现在，如果我画。

32:24
Another point, and the coordinate is two, two, and I don't.
另一个点，坐标是2,2，我不。

32:29
The line between them. I define my line as the line from here to here.
它们之间的线。我把直线定义为从这里到这里的直线。

32:32
Then what you see in your Jesse, that is line, as you find by one one to choose.
然后你在你的杰西身上看到的，那是一条线，当你找到一个一个选择的时候。

32:39
It's just all Linus. You get that as many as you want. It's not infinite though, right? It's as many points as a mix. And you define that line.
你想要多少就有多少，它不是无限的。好吧，尽可能多的毒药和混合，应该找出线

32:50
But they're all straight lengths. They're not co points. They're all straight lengths.
但它们都是直线长度。它们不是0分。他们都是直线的长度。

32:54
Hong Kong.
香港。

33:00
What's a polygraph?
什么是测谎仪?

33:01
How do you think that's a fine?
你觉得这怎么算罚款?

33:06
It's essentially a line that feeds back on itself. It's a close loop.
它本质上是一条自我反馈的直线。这是一个闭环。

33:15
Right? I have another point here. I have another point here. Is this line. Is this line. And it's this line. So essentially it's just a line where the end point and the stock pointer.
对吧?这里还有一点。这里还有一点。是这条线。是这条线。就是这条线。所以本质上它只是一条线，终点和股票指针。

33:27
Listen, listen.
听着,听着。

33:28
Computation. That's all GS. That is right.

计算。这就是GS。没错。

33:33
A series of coordinates, sometimes single coordinates, sometimes a few in one line. Sometimes it's pretty.
一系列坐标，有时是单个坐标，有时是一条直线上的几个坐标。有时它很漂亮。

33:48
So.
所以。

33:48
Very interesting.
很有趣。

33:54
And I asked you guys, what is Singapore, right? The Singapore point or line? The Singapore truly your point or line or polygon.
我问你们，新加坡是什么？新加坡点还是新加坡线？新加坡真的是你的点或线或多边形。

34:05
What is Singapore?
新加坡是什么？

34:08
Is an island, right? There's a physical reality to it, right? What else is Singapore?
是个岛，对吧？这是真实存在的，对吧？新加坡还有什么？

34:17
The nation state, right? So it's a polity, right? It was a social construct of what we think the country is, right? A system of government.
民族国家，对吧？所以这是一个政体，对吧？这是我们对这个国家的社会建

构，对吧？一种政府体制

34:31
With sovereign use of force, ability to produce money.
拥有主权使用武力，有能力创造货币。

34:36
Do to interest students, right? All the traits of a nation state, right? So in this case, a nation state that is tacked to territory.
让学生感兴趣，对吧？这些都是民族国家的特征，对吧？在这种情况下，一个民族国家与领土挂钩。

34:46
Very defined by a character. What else is Singapore?
由一个角色来定义。新加坡还有什么?

34:50
Identity the community is living in it. Well, does that pop up in GS?
认同社区是生活在其中的。这在GS中会出现吗?

35:01
Not really.
不是真的。

35:03
Today, as prioritizes.
今天，作为优先事项。

35:07
Physical space, right? And something that you read in your neo geography reading is that we all know that's very problematic.
物理空间，对吧？你们在新地理读物中读到的是我们都知道这是很有问题的。

35:17

Just let.
让。

35:20
Patterns, experience patterns, memory patterns, communities into something that seems quite objective.
模式，经验模式，记忆模式，群体变成了一些看起来相当客观的东西。

35:29
I think that is ones and zeros.
我想那是1和0。

35:32
So as it's going to be, you need to work against that brain of the thing.
因此，你需要和它的大脑对抗。

35:41
You can still be very useful, though, right? But we need to know that the technology itself is not necessarily on your site.
不过你还是很有用的，对吧？但是我们需要知道技术本身并不一定在你的网站上。

35:49
Okay.
好吧。

35:53
Right?
对吧？

35:53
So.
所以。

35:54

Rest of us? Spector. What is rest and what vector?
我们其他人呢？斯佩克特。静止是什么，矢量是什么？

35:59
So give me an example of arrester data.
给我举个拦阻器数据的例子。

36:08
Something.
某物。

36:08
We work with every day. That's actually rest.
我们每天都在一起工作。这就是休息。

36:10
What is rusty pixels, right?
什么是生锈的像素，对吧？

36:24
If you zoom in into a photograph all the way down, you see.
如果你把照片一直往下放大，你会看到。

36:31
Right?
对吧？

36:31
A photograph is a very good a picture. A digital picture is a very good example of what recipe that is essentially.
一张照片是一张非常好的照片。一张数码照片就是一个很好的例子说明这是什么配方。

36:41
Grits of information. So E.G., it would be easier, sort of like a white.

大量的信息。举个例子，它会更简单，有点像白色。

36:49
This is what?
这是什么?

37:04
Can you put together mix?
你能把这些混合在一起吗?

37:07
That's it, right? I can draw a smiley face right away.
就是这样，对吧？我可以马上画一个笑脸。

37:13
That if it is.
如果是的话。

37:16
Right?
对吧?

37:18
Then your mind kind of sees it as this.
然后你的大脑会把它看成这样。

37:23
That's good.
这很好。

37:27
Why is Rester a good Cathy? What you good at?
为什么雷斯特是个好凯茜？你擅长什么?

37:34

So it's good. You can put value to it, right? What I would say here is this black and his wife.
所以这很好。你可以赋予它价值，对吧？我想说的是这个黑人和他的妻子。

37:54
But you can, instead of one and zero, you can put 0.5, 0.2.
但是你可以，不用1和0，你可以用0.5和0.2。

37:59
Five, 0.2.
5、0.2。

38:01
What it means is that it's progressively lighter and lighter. We can have what we call ingredients, right? We can have things that look really colorful.
它的意思是它逐渐变得越来越轻。我们可以有所谓的配料，对吧？我们可以拥有看起来非常丰富多彩的东西。

38:11
They have ingredient in terms of the color.
它们在颜色方面有成分。

38:14
So it gives us a lot of information, gives us a lot of granularity. So you like heat maps, E.G., that's faster.
它给了我们很多信息，给了我们很多粒度。所以你喜欢热图，比如，这样更快。

38:22
Okay. So that's Rester, which is a grid. What is vector?
好吧。这就是雷斯特，它是一个网格。向量是什么?

38:28
Vector data, right? What makes them vector data?
矢量数据，对吧？是什么使它们成为矢量数据?

38:42
This is also like that one and the other. There's no like reading.
这也像那个和那个。没有比读书更好的了。

38:48
That's the key thing about vector data, right? There's no gradient. There's no halfway in between, right? It's either one or it's zero, right? How is vector data defined? Because for us, we know restor data is essentially the grid, right? And each grid has a value to It.
这是矢量数据的关键，对吧？没有梯度。没有中间地带，对吧？不是1就是0，对吧？矢量数据是如何定义的？因为对我们来说，我们知道还原数据本质上就是网格，对吧？每个网格都有一个值。

39:04
That's how it's defined. So you put a photograph into an Excel, right? All you see, essentially a values.
这就是它的定义。你把一张照片放进Excel，对吧？你所看到的，本质上是一个值。

39:12
One, one, three.
一，一，三。

39:15
Open, one, open street, open to five of that social.
开放，一，开放的街道，开放到五的社会。

39:18
What about back to data? How does that affect?
回到数据上来怎么样？这有什么影响?

39:28
Keeps here.
一直在这里。

39:29
Anyone do dynamics?
有人学过动力学吗?

39:39
Physics, what's effective it? What's a different impact in scale of direction? Direction, right?
物理，什么是有效的？在方向上有什么不同的影响？方向,对吧?

39:48
Direction. It tells you where things are going, how things are related to each other, how they related in space, right? So what Baptists tell you is how far this thing is from here.
方向。它告诉你事物的走向，事物之间的关系，它们在空间中的关系，对吧？所以浸信会教徒告诉你的是这个地方离这里有多远。

40:01
This is defined by an equation. That line is defined by an equation. It's not defined by this point and this point and this point, it's not defined by the infinite points.
这是由一个方程定义的。这条直线是由方程定义的。它不是由这个点，这个点和这个点定义的，它不是由无限个点定义的。

40:11
Is defined by how they are related to each other in this hypothetical space, right? So if I'm going to draw.
是由它们在这个假设空间中的相互关系来定义的，对吧？如果我要画出来。

40:18
This mighty case, right?
这是一个强大的案例，对吧?

40:21
Impact the format. This is essentially coordinate.

影响格式。这就是坐标。

40:29
This is going to be what equals. I can't remember what the formula is called a circle, right? Whatever.
这就等于。我不记得圆的公式了，对吧？无论什么。

40:40
Terrible life, right? If I drew a line, right? Like that, this line is defined by Y was.
糟糕的生活，对吧？如果我画一条线，对吧？像这样，这条直线由Y定义。

40:50
Yes, thanks.
是的,谢谢。

40:53
Is it?
是吗?

40:53
Okay.
好吧。

40:57
I see.
我明白了。

40:57
It doesn't have physical pattern. These are just decisions.
它没有物理模式。这些只是决定。

41:02
What does that mean? It means that it's, I don't want, on zero, right?
这是什么意思？它的意思是，我不想让它在0上，对吧？

41:10
Either this part is filled in or this plant is not filled in.
要么这部分填充，要么这个工厂没有填充。

41:15
Is it on the line, or is it not on the line?
是在线上，还是不在在线上?

41:18
Is this part of the line, or is this one not on the line? Bring this partly, whereas this one is kind of different, right? It can be partly on the line.
这是直线的一部分，还是不在直线上？把这个部分带来，而这个有点不同，对吧？它可能部分处于危险之中。

41:30
Satellite.
卫星。

41:30
And the good thing about this is that you can zoom in however much you want, and you will never get isolated.
这样做的好处是你可以任意放大，而且你永远不会被孤立。

41:41
Because at least the frame of the method.
因为至少是方法的框架。

41:44
Very important for design work.
这对设计工作非常重要。

41:49
It means that if I meet this and I define it by equations, I can blow this up into the size of a building.

这意味着如果我满足这个条件，我用方程定义它，我可以把它放大成一栋楼那么大。

41:59
And it will never look fascinated.
它永远不会让人着迷。

42:01
I know exactly whether something is on the line or not.
我能很清楚地知道是不是有什么事。

42:06
This is why you can put GIS data on the map and zoom in and zoomer and whatever, and it will never lose.
这就是为什么你可以把地理信息系统的数据放在地图上，放大，缩放等等，它永远不会丢失。

42:14
This must always be a great, is it? Yes, there's always a great there's always a great.
这一定很好，对吧？是的，总会有好的总会有好的。

42:26
Ah, but factors have a for the university. We will go into that. But it has to be plotted on some canvas. It needs to have the X and Y axis.
啊，但是因素对大学是有影响的。我们会讲到的。但它必须画在画布上。它需要有X轴和Y轴。

42:38
You need this X one. Otherwise, this one doesn't know what is related to it. This one doesn't know what's related to.
你需要这个X。否则，这个不知道与它相关的是什么。这个不知道和什么有关。

42:45

So there's still a reference. It's just that it's an infinite reference, right? Whereas breasts are constrained by that dimension.
所以还是有参考的。只是它是一个无限参考，对吧？而乳房受到这个尺寸的限制。

42:53
Yes, two dimensional.
是的，二维的。

42:58
Right there is three dimensional geometry is possible. But with.
三维几何是可能的。但随着。

43:04
What we would be going to.
我们将要。

43:06
Which makes something difficult, because if you.
这让事情变得困难，因为如果你。

43:14
You can do this. Okay. So the word three dimension, all right. The word globe, right?
你能做到的。好吧。三维这个词，好的。地球仪这个词，对吧？

43:20
When we try and flatten in two dimensions, we need to make certain tradeoffs. Do you want to be really accurate at a local level, because if you want to be accurate, let's say, a Singapore level, you'll be very inaccurate. Somewhere else on the other side of the Globe.
当我们尝试在二维空间中进行平面化时，我们需要做一些权衡。你想在地方层面做到准确吗，因为如果你想做到准确，比如说在新加坡层面，你就会非常不准确。在地球另一边的某个地方。

43:34
Right? Or do you want to average out so that it's inaccurate everywhere, but to a lower extent? That's the tradeoff.
对吧?或者你想平均一下，这样在任何地方都是不准确的，但在较低的程度上？这就是权衡。

43:43
So there's a difference between a world grid and a local grid. A local grid is very good for stuff immediately in that area. So the Singapore National Grid.
所以世界网格和局部网格是有区别的。局部网格对于该区域的即时内容非常有用。新加坡国家电网。

43:53
Is very good for Singapore, but very bad for Hubway across the world. Very, very inaccurate. On the other side the grid that Google uses.
这对新加坡来说是好事，但对环球航空来说是坏事。非常非常不准确。另一边是b谷歌使用的网格。

44:02
Is fine everywhere. It's not very good for specific places. Just find everywhere. We're spread out. The error everywhere.
到处都很好。它对特定的地方不是很好。到处找。我们散开了。到处都是错误。

44:10
That's one we use because it's global.
我们用这个是因为它是全球性的。

44:13
There's a trade off. You need to make that decide that trade off. You'll find that data set sometimes to toggle between different types of.
这是一种权衡。你需要做出权衡的决定。您会发现数据集有时会在不同类型之间切换。

44:24
Choices for this coordinator frame depend on what you're doing. If you are a government planner and you're planning where the small parts are doing, you don't want any error.
这个协调器框架的选择取决于你在做什么。如果你是一个政府计划者，你在计划小部门的工作，你不想有任何错误。

44:34
Because you might end up tunneling into MIT tunnel, right?
因为你可能会钻进麻省理工的隧道，对吧?

44:39
Then that's not a good thing. So you want super precise for the local and you don't care for the rest of the world because you're not doing anything there.
那就不是一件好事。所以你想要本地的超级精确你不关心世界的其他地方因为你没有在那里做任何事情。

44:45
Whereas if you're just trying to find your way around the city, you don't need that kind of centimeter level decision.
然而，如果你只是想在城市里找到路，你不需要那种厘米级的决定。

44:53
So you're happy to work with, like, ten sentence meter error. That's fine.
所以你很乐意处理十个句子的误差。这很好。

44:57
So again, the question is, what are you using?
问题是，你用的是什么?

45:02
What's an appropriate thing?
什么是合适的?

45:05
An appropriate trade off.
一个适当的权衡。

45:07
Right? So lots of things you think about the whole course that I run on this, so we won't go into it, but you're interested. That's next semester.
对吧?我在这门课上讲的很多东西都是关于这个的，所以我们不会深入讨论，但你们会感兴趣的。那是下学期的事了。

45:19
Okay.
好吧。

45:20
So that's rougher. And factly, two very different types of data. The pieces of them are really, really different. They're good for.
这就比较粗糙了。事实上，这是两种截然不同的数据。它们的每一部分都非常非常不同。它们有好处。

45:31
Different things, right? And you're working in different systems. What kind of, what kind of data policy you work from? You did you guys?
不同的东西，对吧？你在不同的系统中工作。你使用的是哪种数据策略？你们有吗?

45:37
I'm a vector from Ruster and vector.
我是拉斯特的矢量，矢量。

45:42
Can I remember?
我能记起来吗?

45:46

Ship files, right?
船舶档案，对吧?

45:50
HP that's a B that contains points, lines and points.
HP是包含点，线和点的B。

45:59
So Jujing is like a She shot.
所以聚静就像一个她的镜头。

46:05
So you might really focus.
所以你可能真的很专注。

46:11
It's a weird fact.
这是一个奇怪的事实。

46:13
This is, it's me what it's like. So it's essentially like this zero comma one, comma one, comma zero. Comma one, comma.
这就是，这就是我的感觉。它本质上就像这个（0,1），（0,1），（0,0）逗号1，逗号。

46:27
That's what is Yes, right? They are values, and get limited by, separated by commerce.
这就是对的，对吧？它们是价值，被商业所限制，被商业所分离。

46:35
Rest.
休息。

46:44

Dukey, anyone? Guess what A duke is?
提升平车,有人知道吗?猜猜公爵是什么?

46:49
I mean, there are two parts to this, right geo and tip.
我的意思是，这有两部分，右地理和尖端。

46:59
So I wonder what tip I was.
所以我想知道我的小费是多少。

47:07
That's the fifth one.
这是第五个。

47:07
Very good. Thank you. Yes, A tip file is an image file.
很好。谢谢你！是的，提示文件是一种图像文件。

47:17
The tifol is like a jpeck, right? It's like a giff gift. Everyone pronounced it a tiff. A tiff file is a very high resolution.
这个小玩意儿就像个鸟，对吧？就像礼物一样。每个人都说这是口角。tiff文件的分辨率非常高。

47:29
Digital image from usually when people do affection scans, you know, scanning to click.
通常人们做情感扫描时的数字图像，你知道，扫描点击。

47:38
Because tip has very little compression. So your file sizes, like the one I worked on, was something like 500 MP or something.
因为尖端的压缩很小。所以你的文件大小，就像我做的那个，大概是500mp左右。

47:46
All right, for each part, which is very, very big. So most people don't like working tips, because obviously it takes a lot of competition of power, they have to go load it and shift it around, and.
好了，对于每个部分，这是非常非常大的。所以大多数人都不喜欢工作提示，因为很明显这需要很多权力的竞争，他们必须去加载它，然后转移它，而且。

47:58
Do what else and something. All right. So my work with Verji Paginster, something that's more compressed.
做些别的事情。好吧。所以我和Verji Paginster合作的作品，更加压缩。

48:05
Okay. Now, why is this not a tiff, but attuitive?
好吧。为什么这不是口角，而是态度?

48:11
What is the job?
是什么工作?

48:15
Can you guess?
你能猜到吗?

48:23
Link to some coordinates on a map.
链接到地图上的坐标。

48:33
So easier to get there. A dutiff is a tiff that has quantity that is attached to it. So what's happening here? If, let's say, this is your tip.
所以更容易到达那里。关税是一种带有数量的关税。这里发生了什么？假设，这是你的小费。

48:45
Is that this point here has a coordinate link to it. These lines, polygons have a coordinate link to it.
这个点和它有一个坐标链接。这些线，多边形有一个坐标链接。

48:55
To tell you exactly how this would fit onto a jazz system, right? So someone has gone to the trouble of associating all the different coordinates.
来告诉你这是如何适合爵士乐的，对吧？所以有些人把所有不同的坐标联系起来。

49:06
And assigning coordinates each of the points here. So if you load it into QGIS, it will immediately inform.
给这里的每个点分配坐标。因此，如果您将其加载到QGIS中，它将立即通知。

49:14
To your group.
到你的小组。

49:14
Did you?
是吗?

49:16
So it's a tip plus coordinates.
所以这是一个尖端加上坐标。

49:20
It's kind of same here.
这里也是一样的。

49:23
Adjacent, adjacent file is just a data format.
相邻，相邻文件只是一种数据格式。

49:28
But George, jason, there's one that has colleagues.
但是乔治，杰森，有一个人有同事。

49:31
All right, also, I wonder, I know this is a bit heavy, because you're looking for, we're talking about philosophy and ephysiology, and now we don't want Palm formats.
好吧，我也想知道，我知道这有点沉重，因为你正在寻找，我们正在谈论哲学和生理学，现在我们不想要Palm格式。

49:44
Math and math, right? You type notes in equation again.
数学和数学，对吧？你再次在等式中输入注释。

49:49
I want to take very good to go all of this. You mentioned that, you know, if they're of a particular format, if you put it on a GI system, it'll matter.
我想把这一切都带走。你提到过，如果它们是一种特殊的格式，如果你把它放在GI系统上，那就很重要了。

50:03
Automatically. So that means there is some sort of, like industry standard that has been set, right? So how does that come about? Who sets with?
自动。这就意味着有某种行业标准被制定了，对吧？那么这是怎么发生的呢？和谁一起?

50:12
So power formats are not difficult to create. They're just a specific way of defining lots in the past. So you can create a new power format if you want.
因此，创建权力格式并不难。它们只是过去定义地段的一种特殊方式。所以

你可以创建一个新的权力格式，如果你想。

50:21
But you need to specify exactly how a computer should read it, or how a program should read it. Right? The question always is, do you want to use a power format that everybody else is using already.
但是你需要指定计算机应该如何读取它，或者程序应该如何读取它。对吧?问题总是在于，你是否想要使用一种其他人都已经在使用的强大格式。

50:33
Or the other people's programs can use? Or do you want to create something next to you? And why would you need to do that right?
还是别人的程序可以用？或者你想在你身边创造一些东西？你为什么要这么做呢?

50:38
So E.G. checkouts were treated by a company called s Report.
例如，结帐是由一家名为s Report的公司处理的。

50:44
That runs up jazz, up jazz software like HS Where is the main method looking? JAS?
像HS这样的爵士乐软件主要的方法是什么？雅吗?

50:52
LGSIS are very extensive software and.
LGSIS是非常广泛的软件和。

50:58
But governments are rather well use it first. So if you're working in government, you're most likely to use upts.
但政府最好先利用它。所以如果你在政府工作，你最有可能使用up。

51:04
Because it's very easy to produce the stuff that governments need. UJS is

open source, free to use, volunteer run.
因为政府需要的东西很容易生产出来。UJS是开源的，免费使用，志愿者运行。

51:13
Equally powerful, as accurate as not as intuitive.
同样强大，同样准确，但不那么直观。

51:17
And doesn't do how the training programs are up here, stuff right up here.
不像这里的培训项目，这里的东西。

51:24
Makes it very easy for you to learn stuff.
让你学东西很容易。

51:26
Because they have training modules and stuff, and that's.
因为他们有培训模块之类的。

51:30
So this is the economics community.
这就是经济学界。

51:35
Why he would choose to inform us. But the reason why I'm telling you about essay and out of here is because they created the product.
他为什么要告诉我们。但我在这里告诉你们关于论文的原因是因为他们创造了产品。

51:43
Is the industry standard, but people are kind of moving away from it because, you know, it's.
是行业标准，但人们正在远离它，因为，你知道，它。

51:50
Import by a corporation, and so we've moved to its duties, which is an open format.
由一个公司进口，所以我们已经转移到它的职责，这是一个开放的格式。

51:55
But that's the same thing, right? Different certifications a little bit, but ultimately, your adjacents are more accessible.
但这是一样的，对吧？不同的认证，但最终，你的邻接更容易接近。

52:04
From what opens us easily read. And now these programs are very happy to use your business as well.
从容易打开我们的东西中读出。现在这些程序也很乐意使用您的业务。

52:10
So additional problem.
另外一个问题。

52:13
Correct.
正确的。

52:18
Okay.
好吧。

52:18
That's a question, right? Because the ways that we think about data and how, what might you choose to store and has all sorts of ramifications in terms of who might be good access and use them.
这是个问题，对吧？因为我们思考数据的方式和方式，你可能选择存储什么，以及在谁可能很好地访问和使用它们方面有各种各样的后果。

52:30

I also had a question about projections, math projections, because now the last week, there's in this whole, like debate in the UN about the.
我也有一个关于预测的问题，数学预测，因为上周，在联合国有一场关于。

52:43
For Ketter projection versus the equal Earth of area, right? So how is that the adoption or the rejection of these different projections?
对于凯特投影和同等面积的地球，对吧？那么，对这些不同预测的接受或拒绝是怎样的呢?

52:52
Are going to affect this sort of like open source work. But some people would use certain projections. Some people would prefer.
会影响到像开源这样的工作。但有些人会用特定的投影。有些人更喜欢。

52:59
The answer is it doesn't really affect the chairs work, because there are two levels to protect that.
答案是它不会真正影响椅子的工作，因为有两个层面来保护它。

53:09
Okay, let's go into this. So we have a three dimensional world, right? And so what?
好了，我们开始吧。我们有一个三维世界，对吧？那又怎样?

53:18
What we try to do is we try to create a mathematical model that probably works.
我们要做的是试着建立一个可能有效的数学模型。

53:24
Or how the world is, how the planet is not.
或者说世界是怎样的，地球又是怎样的。

53:29

It bulges right around the equator because it's spinning, so it's a bit flat. That opposes the geometry of it is not even because of mountain ranges and all that. And.
它在赤道附近凸起，因为它在旋转，所以有点平。这与它的几何形状相反，甚至不是因为山脉之类的。和。

53:40
Of gravity and all that stuff, right? So what we're doing is mathematicians and map people have created a mathematical model which will be able to.
重力什么的，对吧？所以我们所做的是数学家和地图专家创造了一个数学模型，它将能够。

53:50
Recreate the shape of the globe.
重现地球的形状。

53:53
Of course, there are some tradeoffs right now.
当然，现在有一些权衡。

53:58
The key thing to note is that the mathematical model is still treaty. It's still in three dimensions. It's only.
需要注意的关键是，数学模型仍然是条约。它仍然是三维的。只有。

54:06
When you produce it and you visualize it, then it comes to.
当你产生它并想象它的时候，它就来了。

54:10
So coordinates still belong on a treaty.
所以坐标仍然属于条约。

54:13
So it doesn't matter which projection means, whether it's Mecket or equal

or because that question only comes in when you are transmitting it into the author.
所以投影意味着什么并不重要，不管是Mecket还是equal，因为只有当你把它传递给作者时，这个问题才会出现。

54:21
So that's why the answer is it doesn't actually matter.
这就是为什么答案是无所谓的原因。

54:26
Let me, maybe I can brush it up.
让我，也许我可以复习一下。

54:29
So we've got a world, right? Actual.
所以我们有了一个世界，对吧？实际。

54:35
And then we've got a model.
然后我们有了一个模型。

54:39
Right? This is go three D. This is where your coordinates did.
对吧?这是三维，这是你的坐标。

54:48
So coordinates refer to this treaty model.
坐标指的是这个条约模型。

54:52
Then when we actually see it on a screen, or when we print out a map, it becomes a two D.
然后当我们在屏幕上看到它时，或者当我们打印出地图时，它就变成了二维。

55:02
And this is where we call it a projection. This is when the projection happens. So this is the Makita.
这就是我们称之为投影的地方。这是投影发生的时候。这是Makita。

55:12
The equal of.
等于。

55:14
Or whatever, pulla, right?
或者别的什么，拉，对吧?

55:21
So the translation, what happens here? But all the GS work that you're working with, all the geo data they're working with actually still belongs in the treaty model.
那么平移，这里发生了什么？但是所有你正在处理的GS工作，他们正在处理的所有地理数据实际上仍然属于条约模型。

55:32
That's why it doesn't matter. The only question is, when you put it up there into the world, when you're putting on the screen, where you put it on the Internet, and you print it out.
这就是为什么它不重要。唯一的问题是，当你把它放到这个世界上的时候，当你把它放到屏幕上的时候，当你把它放到互联网上的时候，你把它打印出来。

55:41
You have to make this choice. I want to projection.
你必须做出选择。我想要投影。

55:44
And it is, well, that issues coming.
问题来了。

55:46
Right? Because Mekita, if you're not aware, with the debate, the Mekita map projection makes equatorial stuff like Africa looks smaller.
对吧?因为Mekita，如果你不知道，在这场辩论中，Mekita地图投影使得赤道地区，比如非洲，看起来更小。

55:56
And stuff like Greenland, much bigger. Europe, much bigger. North America looks much bigger, right? Remember with how much tradeoffs, right? When you want to convert something of treaty into Tuddy to the make tradeoffs, the tradeoffs they made was that the stuff up there will be made Bigger.
像格陵兰岛，要大得多。欧洲要大得多。北美看起来要大得多，对吧？还记得有多少权衡吗？当你想把条约中的一些东西转换成图迪来进行权衡时，他们所做的权衡是上面的东西会变得更大。

56:11
And I start below, but comparatively look smaller.
我从下面开始，但相对较小。

56:13
That is titled to Colonialism, right? About what is more important, what's more prominent, right? And when people look at maths, they see it in a classroom.
这就是殖民主义，对吧？关于什么更重要，什么更突出，对吧？当人们看到数学时，他们看到的是在教室里。

56:23
Right? I think that is the word. It's not right. That's a sixteenth century decision, with a fifteenth Makita, right, to draw the world, to project the world. And then.
对吧?我想就是这个词。这是不对的。这是十六世纪的决定，十五世纪的Makita，绘制世界，投射世界。然后。

56:35
So getting confusion. So I'm saying that, let's say you're you, right. I take a photo of you, and it is on the photograph.
这很混乱。所以我说，假设你是你，对吧。我给你拍了一张照片，它在照片上。

56:44
So the vector, or the restor data is able to map you. I'm able to map, represent you in that photograph.
所以矢量，或者说还原器数据可以映射你。我能在那张照片中描绘出你。

56:53
True data, but it doesn't mean that it is, that data is used correct, so it is too useful like So the breaking down of data.
真实的数据，但这并不意味着它是，数据被正确使用，所以它太有用了，所以数据的分解。

57:05
Into all these formats are still useful when I want to represent you on a photograph? Yes.
当我想在一张照片上代表你的时候，所有这些格式仍然有用吗？是。

57:11
But useful for certain things, and not useful evidence. So e.g., you take a picture of me from the front, you don't have any information about me from the back.
但对某些事情有用，而不是有用的证据。例如，你从正面拍了一张我的照片，你不知道我背后的任何信息。

57:19
Right? You don't have information about, I don't know, between my foot and the floor, but I can actually represent your head to toe.
对吧?你不知道，我不知道，我的脚和地板之间的距离，但我实际上可以代表你的从头到脚。

57:28
That's what I want to achieve then, yes. The question then is accuracy.
这就是我想要实现的，是的。接下来的问题是准确性。

57:36
All right, is this a good enough proxy of me for your purposes?
好了，这能代表你想要的我吗?

57:42
That's a very funny story. So, say Yucchin, actually, we do not know whether sa yucin had a.
这是一个非常有趣的故事。所以说，尤金，实际上，我们不知道尤金是否有。

57:52
Had a queue or not. So that that's the picture, right for Chinese who came over, and he's great, good grandson who I was talking to, and if she asked him.
有没有排队。这就是照片，对那些来的中国人来说，他是我的好孙子，我和他聊天，如果她问他。

58:02
And unfortunately, all the photos that they have are whipped off the head, so he's wearing a hit piece, and they're of the wrong angle, so you can actually tell away against a Q or not. So, so it's one of those things where you can have a lot of photos of a person.
不幸的是，他们手上所有的照片都是从他头上剪下来的，所以他戴的是假发，而且角度不对，所以你可以根据Q来判断。这是其中一个你可以有很多一个人的照片的东西。

58:16
You're still missing that you don't have to write Peter. So how would you?
你还是很想念你不用给彼得写信。那你怎么做呢?

58:25

Get around that. The good question.
避开这个问题。问得好。

58:28
I mean, he has taught his own family archives, but he doesn't.
我是说，他教过自己的家族档案，但他没有。

58:33
At least not from that the archives.
至少从档案里找不到。

58:36
So that's okay. That wouldn't come from the photographs.
所以没关系。这不是来自照片。

58:40
Come from family stories, maybe from the Shadow Against the Screen.
可能来自家庭故事，也可能来自《屏风上的影子》。

58:47
I mean, some people do this for art analysis, right? We do X ray spectography to look at artwork and see, actually, when the paint was laid.
我是说，有些人这样做是为了艺术分析，对吧？我们用X射线光谱法来观察艺术品，看看油漆是什么时候涂上的。

59:00
What were the initial brass strokes? Because sometimes painters brush over their initial ones. They might do it in a particular way. Then they decide, actually, this doesn't look good, and they decide to do another way.
最初的黄铜笔触是什么？因为有时画家会刷掉他们最初的画。他们可能会用一种特殊的方式来做。然后他们决定，实际上，这看起来不太好，他们决定用另一种方法。

59:09

So they draft first, and then they retract, you put something on top of it. So X ray cryptography can help us sort of see this like working progress. Sometimes it helps us with authorship.
所以他们先起草，然后收回，你在上面放一些东西。所以X射线密码学可以帮助我们看到这种工作的进展。有时它能帮助我们确定作者身份。

59:19
Right? So as a team, whether something is real, painters work in very specific ways or they do very specific methods. And you can.
对吧?所以作为一个团队，无论某样东西是真的，画家们都以非常具体的方式工作，或者他们用非常具体的方法。你可以。

59:27
And see from under, whether it was actually them or not.
从下面看，到底是不是他们。

59:30
But that's also digital history, in a way, right? That's looking at sort of different methods to answer historical questions.
但在某种程度上，这也是数字历史，对吧？就是用不同的方法来回答历史问题。

59:39
Great.
好了。

59:43
Okay. So that's GS, and that's the data that you deal with in GS. And some of that, there's some logical issues with presenting.
好吧。这是GS，这是在GS中处理的数据。其中的一些，在陈述时存在一些逻辑问题。

59:56
Representing a treaty space in two D right?
表示二维的条约空间，对吧?

01:00:00
Factors of colonialism, right? Come in. Factors of flattening, right? How do you represent? How can you truly represent?
殖民主义的因素，对吧？进来。平坦化的因素，对吧？你是如何代表的？你怎样才能真正地表现？

01:00:11
A living, breathing, empty, or communities, and all its complexity using just points like the polygons.
一个有生命的，有呼吸的，空的，或社区，以及它所有的复杂性都只是使用像多边形这样的点。

01:00:18
Not right. So how do we, as historians, sort of bring that contextually or back right? And that's what New Geography is also trying to share similar goals.
不正确的。那么，作为历史学家，我们该如何，把这些联系起来呢？这也是《新地理》也在尝试分享相似目标的原因。

01:00:29
Right? In terms of making jazz represent reality in the ways that we see it.
对吧?让爵士乐以我们所看到的方式表现现实。

01:00:36
What about spatial history, right?
那么空间历史呢?

01:00:42
I think you guys read Richard White. I think Richard White is the leader of I don't think it's surrounding the model. The speech is really that at Stanford.
我想你们读过理查德·怀特的书。我认为理查德·怀特是我不认为这是围绕模型的领导者。斯坦福的演讲就是这样。

01:00:52
Right? And he kind of recounts what it means by bringing in a special dimension to history, right?
对吧?他通过引入一个特殊的历史维度来叙述它的意义，对吧?

01:00:59
It brings up new questions, right? It brings us new directions in terms of what we want to investigate as historians, right?
这带来了新的问题，对吧？它为我们作为历史学家的研究提供了新的方向，对吧?

01:01:09
Um.
嗯。

01:01:10
What kinds of questions, though, need a special dimension. What do you think?
但是，什么样的问题需要一个特殊的维度呢？你觉得呢?

01:01:18
When you're looking at questions about mobility, movement, migration? Well, very big questions today, right? We talk about global history.
当你看到关于流动性，迁徙，移民的问题时？今天的问题都很重要，对吧?我们谈论全球历史。

01:01:32
We talk about transnational history. Movement is a key part of the story. What's your life? Which of I still like, okay, very good.
我们谈论跨国历史。运动是故事的关键部分。你的生活是什么？我还是喜欢哪一个，很好。

01:01:42
He's retired right from the.
他已经从。

01:01:45
Okay, I'll probably read that. I don't know whether his history lab is still around me. Maybe not.
好吧，我可能会读。我不知道他的历史实验室是否还在我身边。也许不是。

01:01:53
I think it, I think it's shocked. But yes, migration history, transnational history, right? Migration movement is a key part.
我觉得，我觉得它很震惊。但是，是的，移民史，跨国史，对吧？移民运动是一个关键部分。

01:02:04
Of this story, right to you guys. About the.
这个故事的一部分，给你们。有关。

01:02:10
Tooth isotop, right with the oxygen isotop and beef.
牙齿同位素，氧同位素和牛肉。

01:02:17
Do you remember?
你还记得吗?

01:02:18
The story where we were looking at archaeological remains of bones of people and their teeth, and where they grew up, right? And that's determined by the isotop of oxygen and their Teeth.
在这个故事中，我们看到了人类骨骼和牙齿的考古遗迹，以及他们在哪里长大，对吧？这是由氧的同位素和它们的牙齿决定的。

01:02:29
And that gives us a sense that they were born in the Mediterranean, and they moved during their lifetime to you.
这给我们一种感觉，他们出生在地中海，在他们的一生中搬到你这里。

01:02:35
The British House, right? And so there's a special component there that tells us a little bit about how they physically moved, but also goods.
英国议会，对吧？这里有一个特殊的部分告诉我们他们是如何移动的，还有货物。

01:02:45
Right? Ideas, right? Not many things that move, not just people.
对吧?的想法,对吗?不会动的东西不多，不只是人。

01:02:52
You can look at back here, you know, animals, flora and fauna, those move.
你可以看看这里，你知道，动物，植物群和动物群，它们会移动。

01:03:00
Right, martin, you work in natural history. Specimens move right. Specimens are collected. They are collated.
好吧，马丁，你是自然历史系的。标本向右移动。采集标本。它们是经过整理的。

01:03:10
Archive.
档案。

01:03:12
So Mike, mapping some of that movement, tell us something about wider transit history, population demographics, right? Questions of nationality.
迈克，给我们描绘一下这些运动，告诉我们一些更广泛的交通历史，人口统计数据，对吧？国籍问题。

01:03:25
Questions of race, community, and religion.
种族、社区和宗教问题。

01:03:29
Right? Questions of labor and gender. I'm just teaching about the marches and the Amas and Singapore divative, right? So these are women, unmarried women from the pro Delta region in Singapore.
对吧?劳动和性别的问题。我只是在教游行，阿玛斯和新加坡的婚礼，对吧?这些是来自新加坡前三角洲地区的未婚女性。

01:03:41
To look at domestic workers, help raise children, help clean the house right and.
看看家政工人，帮忙抚养孩子，帮忙打扫房子对不对。

01:03:50
These are also geographically displaced populations, right? They are moving forward, and they can, because there was no restriction on women migrating to Singapore.
这些也是地理上流离失所的人，对吧？她们正在向前发展，而且她们能够向前发展，因为过去对女性移民新加坡没有任何限制。

01:03:59
So it should come men.
所以应该是男人。

01:04:01
So the gender dimension to this legal migration flow.
这是合法移民流动的性别层面。

01:04:07
It's my best meeting key policy in foreign industry.
这是我遇到的最好的外企政策。

01:04:11
Okay. So special history, any other examples or special history that jumps out at you when you're reading it?
好吧。特殊历史，你们在读的时候有没有其他的例子或者特殊历史?

01:04:20
For the forms of history.
对于历史的形式。

01:04:26
A benefit from a special dimension.
一个特殊维度的好处。

01:04:28
Symbolism.
象征意义。

01:04:35
The flight.
飞行。

01:04:37
Say, school or religious base of worship or your parks. That's how people like to view, seeing how they live in the area.
比如说，学校或宗教崇拜基地或你的公园。人们喜欢这样看，看看他们在这个地区是如何生活的。

01:04:48
So the special relationships between things, right? Not just movement, but just proximity.
事物之间的特殊关系，对吧？不仅仅是移动，还有接近。

01:04:57
Co location right is very useful for helping us surface relationships that we may not see in text.
共同定位权对于帮助我们呈现文本中看不到的关系非常有用。

01:05:05
Right? Um.

对吧?嗯。

01:05:08
Here's an example. So when I was in Cambridge, I was looking at some of the ordinances of what it means to be a student there.
这里有一个例子。所以当我在剑桥的时候，我看了一些关于在那里成为一名学生意味着什么的条例。

01:05:16
And it turns out to be a student. To be an active student, you need to live within hearing radius of the church bells, the single town.
结果是一个学生。要成为一名活跃的学生，你需要住在教堂钟声能听到的范围内。

01:05:25
That's how it means it's defined to be in resident, so you must be able to hear the bells. So there's a certain.
这就是它被定义为常驻的意思，所以你必须能听到钟声。所以有一定的。

01:05:34
Distance, right?
距离,对吧?

01:05:37
Does this definition? It's not a kilometer definition, oral definition. So then you can actually look at the geographical reality of that.
这个定义正确吗？这不是一公里的定义，不是口头的定义。这样你就可以看到它的地理现实。

01:05:47
Doesn't mean because if you're blocked by something, right? The sun doesn't travel there. If you have a forest in between you and the bells, right? That muffles the noise, and you cannot hear anymore.
并不是说如果你被什么东西挡住了，对吧？太阳不会到那里去。如果你和钟声之间有一片森林，对吧？这样就消音了，你就听不见了。

01:05:58
This is a very interesting, special puzzle.
这是一个非常有趣、特别的谜题。

01:06:03
How does it mean to be?
是什么意思?

01:06:05
How about Singapore? It's a point, or it's a polygon, right? What does it mean to be in Singapore, right?
新加坡怎么样？它是一个点，或者是一个多边形，对吧？在新加坡意味着什么，对吧?

01:06:12
You need to be at the center point of Singapore in Singapore? Or do you just need to be within the polygon? Is that being in Singapore? If you are in Florida, within the in Singapore.
你需要站在新加坡的中心吗？还是只需要在多边形内？那是在新加坡吗？如果你在佛罗里达，在新加坡。

01:06:22
Right? Or is that not the mainland, and therefore not releasing a lot different?
对吧?还是说中国大陆不是这样，所以发行的电影不太一样?

01:06:29
Simple, right?
简单,是吧?

01:06:30
It's in a book somewhere where you can access the MRC, because you can do that on an off side list.
它在一本书的某个地方，你可以访问MRC，因为你可以在一个侧面列表上

做。

01:06:34
Singapore, where you have reception of Singapore network, right? Like?
新加坡，那里有新加坡网络的接收，对吧？像什么？

01:06:39
Think, tell, right? But doesn't mean possible. Asia is Singapore.
想想，说说，对吧？但这并不意味着可能。亚洲是新加坡。

01:06:43
Right? So how do you define that? What is the relationship? What defines that relationship? Speech, relationship?
对吧?那你怎么定义呢？这是什么关系？是什么定义了这种关系？演讲中,关系？

01:06:53
A general talking about, sort of collocation, just being able to see something down the street, right if you are. Something that we do in temporality studies is.
一般来说，这是一种搭配，只要能看到街上的东西，对吧。我们在时间性研究中做的事情是。

01:07:03
Where do you see clocks? Where are the public clocks? How do people know what time it is? Right? And so people use, rich people use to fund public clocks so that we can all share the same sense of what time it is.
你在哪里看到时钟？公共时钟在哪里？人们怎么知道现在是几点？对吧?所以人们，富人会资助公共时钟，这样我们就能共享相同的时间感。

01:07:17
You put it in a town square, you put it in public parks. You know, some people can go change their watches. According to that if you never watch them, you have to go there and see.
你把它放在城市广场上，你把它放在公园里。有些人可以去换手表。如果你

从来不看，你必须去那里看看。

01:07:27
Right? So do people have access to the time?
对吧?大家有时间吗?

01:07:31
Special fashion. Do people have access to light or natural gas? To body stove or sewers?
特别时尚。人们能用上电灯或天然气吗？去炉子或下水道?

01:07:42
Right? One of the key examples of public health is John snow's Cholera met. Does anyone know that one?
对吧?公共卫生的一个重要例子是约翰·斯诺的霍乱。有人知道这个吗?

01:07:51
So back in the day, right? Not to not to be an academic, but back in the day.
回到过去，对吧？不是说我不是个学者，而是说过去的事。

01:07:59
People used to think that my asthma, right disease is because of bad air. My asthma right. So disease is transmitted by bad air.
人们过去认为我的哮喘是由于空气不好造成的。我的哮喘，对。所以疾病是通过恶劣的空气传播的。

01:08:08
You're in a bad air for long enough, you get a disease in U type, right? And then microbial theory, right? That disease comes from microbes, right? Like microorganisms.
你在恶劣的空气中待的时间足够长，你就会患上U型疾病，对吧？然后是微生物理论？这种疾病来自微生物，对吧？如微生物。

01:08:19

Was kind of a fringe, cheery, right? But John Snow is a physician, very interesting man. He walked from Edinburgh to London to attend medical school.
有点像刘海，很有趣，对吧？但是约翰·斯诺是个医生，非常有趣的人。他从爱丁堡步行到伦敦上医学院。

01:08:27
Crazy distance. But he drew a map. He mapped up all the cases of cholera, which is a waterborne disease. So the microbes exist in water.
疯狂的距离。但是他画了一张地图。霍乱是一种水传播疾病，他把所有霍乱病例绘制成地图。所以微生物存在于水中。

01:08:38
He drew a map of London, soho, and he mapped out where all the cholera pieces were. And then what he did was he juxtapose it against where the water pumps are. Where do people get there?
他画了一张伦敦苏荷区的地图，并标出了霍乱的各个部位。然后他把它和水泵的位置并置。人们从哪里到达那里?

01:08:49
Water from And it turns out all the cases are clustered around one pump.
所有的箱子都聚集在一个水泵周围。

01:08:54
Not around any of the other pumps. All the people who were getting cholera, kim took their water from the same pump by jux, supposing the location of the water.
其他泵附近都没有。所有得了霍乱的人，他都从同一个水泵里取水，假设取水的位置。

01:09:06
And the cases. You can show a link to these two things.
还有箱子。你可以显示这两个东西的链接。

01:09:10

And this is why they teach you in public health, one on one. This is the birth of the discipline, right? Evidence, looking at relationships, testing factors.
这就是为什么他们会在公共卫生课上一对一地教你。这就是这门学科的诞生，对吧？证据，观察关系，测试因素。

01:09:21
Modern medicines, like corporate, various friends.
现代医药，像企业，各种朋友。

01:09:26
That's Grs. If you walk past my door, you'll see that's the map. That's johnson's map.
Grs。如果你走过我的门，你会看到那是地图。这是约翰逊的地图。

01:09:34
It's the chlorine.
是氯。

01:09:38
A very interesting historical example. You can still do this thing, and people still do that today, right? People working in public planning, right? Think about if I was going to build this thing here.
一个非常有趣的历史例子。你现在还可以这么做，人们现在还在这么做，对吧？从事公共规划工作的人，对吧？想想如果我要在这里建造这个东西。

01:09:51
Would it affect the sidelines of something? Right? Would it block the forest? Would it block we've been thinking about bird migration patterns.
它会影响到场边的事情吗？对吧?它会挡住森林吗？我们一直在思考鸟类的迁徙模式。

01:10:01
Right? If I would build a multi storage degree block here, would it affect the migratory patterns of the books that we're using? Was a treatment.
对吧?如果我在这里建立一个多存储度的块，它会影响我们正在使用的书的迁

移模式吗？是一种治疗方法。

01:10:09
Right? Or do we need to sort of skill it in such a way that it provides sort of an open space for them?
对吧?或者我们是否需要某种技能，以某种方式为他们提供某种开放空间?

01:10:14
With your keeping taps on what's happening in the capital area, create a sudden waterfront, right? These are the questions of planning questions that we're thinking about.
你一直关注着首都地区的动向，突然创造了一个滨水区，对吧？这些都是我们正在思考的规划问题。

01:10:24
Somebody's thinking about that.
有人在思考这个问题。

01:10:26
Right?
对吧?

01:10:28
So speciality plays a very big part in understanding society, understanding design, why things happen.
所以专业在理解社会，理解设计，理解事物发生的原因方面起着非常重要的作用。

01:10:38
And Richard white's argument is that we don't think about it that much, even though we live it.
理查德·怀特的观点是，尽管我们生活在其中，但我们对它的思考并不多。

01:10:43
Right? And so jazz can and special history more generally. Or South Jazz

itself brings all that back into the conversation.
对吧?因此，爵士乐可以更广泛地影响特殊历史。或者南爵士本身把所有这些都带回了对话中。

01:10:52
Question stops. But you work in spatial history, are you already doing spatial history? You just haven't been to somebody.
停止的问题。但是你研究的是空间历史，你已经在研究空间历史了吗？你只是没去过别人那里。

01:11:01
Some of you might be there, right?
你们中的一些人可能在那里，对吧?

01:11:08
Okay. Are we good to go? We can proceed.
好吧。我们可以走了吗？我们可以继续了。

01:11:14
All right, cool. Thank you.
好吧，酷。谢谢你！

01:11:20
This is before we take a break, neurography, right? We've been working with this idea, but I want to just.
这是在我们休息之前，神经造影，对吧？我们一直在想这个主意，但我想。

01:11:33
To make sure we go through some of these concepts here. But New Geography is really about opening things up, the curation up to.
为了确保我们在这里讨论一些概念。但《新地理》真正要做的是开放，是策展。

01:11:43
A wider public, right thinking about how communities can make sense,

place around them and incorporate the level of contexts that we assist. Orange Love.
一个更广泛的公众，正确的思考如何让社区变得有意义，在他们周围放置，并结合我们所帮助的背景水平。橙色的爱。

01:11:54
Very into geographically.
非常喜欢地理。

01:11:56
How do people actually see the joke around them, right?
人们是如何看待他们周围的笑话的?

01:12:01
What do they notice? Because you see many things, right?
他们注意到了什么？因为你看到了很多东西，对吧?

01:12:06
The body receives also the signals. But you'll need notice certain things, certain things stand up to you, and what stands up to you.
身体也接收到这些信号。但你需要注意某些事情，某些事情经得起你的考验，以及什么经得起你的考验。

01:12:13
We all know that an object can be seen in multiple ways by multiple people. They have different resonances in different things, for different communities, for people of different ages.
我们都知道，一个物体可以被不同的人以不同的方式看到。它们对不同的事物、不同的群体、不同年龄的人有不同的共鸣。

01:12:25
You have to come backgrounds, right? And JS will just tell you, this is a place, this is, you know, this is a line, and this is a polygon, right? The school is a polygon.
你必须有背景，对吧？JS会告诉你，这是一个位置，这是一条直线，这是一

个多边形，对吧？学校是一个多边形。

01:12:36
Right? But then you all have different experiences of what school is, and your geography tries to capture some of that.
对吧?但是你们对学校都有不同的经历，你们的地理位置试图捕捉其中的一些。

01:12:44
Extra contextual layers.
额外的上下文层。

01:12:54
And so.
所以。

01:12:55
Just thinking about, sort of correspondent here, of truth. I know the article goes a little bit into epistemology and the nature of truth itself, right? Which is kind of important because that's what we've been working with this whole time.
我只是在想，在这里，关于真相。我知道这篇文章涉及到了认识论和真理本身的本质，对吧？这很重要，因为这是我们一直在研究的问题。

01:13:07
Right when I'm asking you guys, what does it mean to represent something on GS, on a digital space that you're actually asking about truth?
就在我问你们的时候，在GS上，在数字空间上表示一个东西是什么意思，你们实际上是在问真相?

01:13:16
Because it retain its true value when we turn it into points, lines and polygons.
因为当我们把它变成点、线和多边形时，它仍然保持着它的真实价值。

01:13:23
Great.
好了。

01:13:23
So we've got a tree. The trees also have this building. Okay.
我们有一棵树。树木也有这样的建筑。好吧。

01:13:31
Then one, right? A hundred year old tree, a massive, I don't know, bunyan tree or something, right? A Templesu tree that exists in Youtown.
然后是1，对吧？一棵百年老的树，一棵巨大的，我不知道，兔子树之类的，对吧？一棵存在于Youtown的神庙树。

01:13:40
Right? We can put that ring's view on the GER system.
对吧?我们可以把这个环的观点放到GER系统上。

01:13:44
Is that all there is to that tree?
这就是那棵树的全部吗?

01:13:47
We're not right. It has emotional resonances, the relationships that have to do with the three memories, types of rich right?
我们错了。它有情感共鸣，与三种记忆有关的关系，丰富的类型，对吧?

01:14:00
Um, might end up on your dollar note, right?
可能会出现在你的美钞上，对吧?

01:14:06
One in the Potentic Gardens.
一个在潜势花园。

01:14:07
So it's an icono symbol.
所以这是一个图标符号。

01:14:11
But on the chairs, it's a point. So how do we work against that, against the reductionist element?
但在椅子上，它是一个点。那么我们该如何反驳，反驳简化论者呢?

01:14:21
Of the steel system.
钢铁系统的。

01:14:23
Not easy.
不容易的。

01:14:27
But of course, then what you're saying is, you know what, if we get to the negatives of truth? Does anyone remember what neighborhoods of truth?
当然，你的意思是，你知道吗，如果我们谈到真理的否定？有人记得真理的邻域是什么吗?

01:14:36
Very interesting term.
非常有趣的术语。

01:14:38
So we've got, course, course, momentary of truth. What's analogy of truth?
所以我们有，当然，当然，瞬间的真相。什么是真理的类比?

01:14:49
I remember.
我记得。

01:14:51
I want to T, okay. Okay.
我想T，好的。好吧。

01:15:02
Correspondentary, two editors.
记者，两位编辑。

01:15:06
Coherent, right? Whether the populations make sense together.
连贯的,对吧?这些人口是否有意义。

01:15:18
That they're coherent. They might be antider to reality by at least they're coherent.
它们是连贯的。它们可能相对于现实，至少它们是连贯的。

01:15:23
And the last one, which I don't think the article goes through, but pragmatic.
最后一点，我认为文章没有讲到，但很实用。

01:15:29
Something is useful, then.
那么，有些东西是有用的。

01:15:32
Jas is a very correspondent area. What is there? Is it there? Is it there? Okay. That's reality.
日本是一个非常相似的地区。有什么？它在那里吗？它在那里吗？好吧。这是现实。

01:15:42
And something that this article talks about is that what we can do is we can maybe focus less on the correspondence and look at the coherence.

这篇文章讨论的是我们可以做的是我们可以少关注对应而关注连贯。

01:15:54
Kind of truth, right? A community makes sense of this in this possible way, doesn't have to necessary correspond to what other people think of the same place.
这是事实，对吧？一个社区以这种可能的方式来理解这一点，并不一定要与其他人对同一地方的看法相对应。

01:16:04
As long as within themselves, they are coherent. And that's why they call it neighborhoods of truth, right? They kind of exist in and of themselves than neighborhoods.
只要它们内部是一致的。这就是为什么他们称它为真理的社区，对吧？比起社区，他们更像是存在于自身之中。

01:16:14
It makes sense by themselves.
它本身是有道理的。

01:16:19
Is that the thing?
是这样吗?

01:16:22
Who thinks is a good thing to enable it's a truth. Who thinks it's a bad thing to enable it? That green if you think is a good thing, and red if you think that we shouldn't actually have neighborhoods on truth?
谁认为是好事，就使它成为真理。谁认为启用它是件坏事？绿色表示你认为是好事，红色表示你认为我们不应该在真相上建立邻里关系?

01:16:34
Half half what you're living the truth. It depends on what the truth is meant for, right?
一半一半你活在现实中。这取决于真相的意义，对吧?

01:16:44
Okay. What's a what's a good case? And what's a bad case?
好吧。什么是好案例？什么是坏情况呢?

01:16:48
What's a good neighborhood?
什么是好的社区?

01:16:58
If you're trying to understand how different communities see, let's say evaluatery, then you will need these neighborhoods of truth. You have to see what different.
如果你想了解不同群体的看法，比如评估，那么你就需要这些真理社区。你得看看有什么不同。

01:17:13
There may be differences. You may be looking for, the differences. You will want to see variations in people's perceptions of a tree. Okay. So the comparative aspect of.
可能会有不同。你可能在寻找不同之处。你会想看到人们对一棵树的不同看法。好吧。所以比较方面。

01:17:24
Different neighbors of truth. It's a good thing because it enriches our ability to describe what's happening on it.
真理的不同邻居。这是一件好事，因为它丰富了我们描述地球上发生的事情的能力。

01:17:31
Oh, okay.
哦,好的。

01:17:32
What's bad?

糟糕的是什么?

01:17:39
Equitum.
Equitum。

01:17:40
Echo chambers. So it could be possible that, you know, we are creating a loop, right?
回音室。所以有可能，你知道，我们正在创建一个循环，对吧?

01:17:47
Is that the property of communities in general.
这是群体的一般属性吗?

01:17:53
We think about the theories of culture and cultural change, right? The culture develops as a American copy of being the group in the community.
我们考虑过文化和文化变迁的理论，对吧？这种文化是作为社区群体的美国复制而发展起来的。

01:18:06
And cultures, norms and ideas and ideologies that we share and we disseminate right.
以及我们分享和传播的文化、规范、思想和意识形态。

01:18:14
Sharing culture is essentially makes you have any good This is someone.
分享文化本质上是让你有任何好的这是有人。

01:18:22
Doesn't share in the way that's an accurate chamber.
不能分享这是一个精确的枪膛。

01:18:26

Could you cut yourself off?
你能打断一下自己吗?

01:18:29
So it's interesting because, of course, these are all cut several weeks, right?
这很有趣，因为，当然，这些都是在几周内切割的，对吧?

01:18:43
Um.
嗯。

01:18:43
People in new geography wouldn't say that, E.G., if it's not tied specifically to specific parts of geography, it's invalid.
新地理学的人不会这么说，例如，如果它没有与特定的地理区域联系在一起，它就是无效的。

01:18:54
Right? It can be sort of an imagine geography. It can be a relational geography. So, E.G.?
对吧?它可以是一种想象的地理。它可以是关系地理。那么,如?

01:19:01
Um.
嗯。

01:19:02
There are very few examples of indigence maps in America.
在美国很少有贫穷地图的例子。

01:19:07
The most famous map is a map which we call the Cophova dmap. And what we see are squares.
最著名的地图是一张我们称之为科霍瓦地图的地图。我们看到的是正方形。

01:19:19
Triangles, square.
三角形、正方形。

01:19:22
Here, and then there's a line here. There's a line here, there's line here.
这里，这里有一条线。这里有条线，这里有条线。

01:19:29
And this is on the BS skin.
这是在BS皮肤上。

01:19:33
Kind of our primary example of an indigenous stopping record. Now you look at it and you'll be like.
这是土著停止录音的主要例子。现在你看着它，你会说。

01:19:43
West geography, right?
西方地理，对吧?

01:19:47
And it turns out that this doesn't have anything geography. This tells you that this person is more closely related to that person and that person is more closely related.
事实证明，这与地理无关。这告诉你这个人和那个人关系更近，那个人关系更近。

01:19:58
Or this case, these are the people that these are communities talk to each other, and these communities talk to each other. So it's much more like a relationship network.
或者在这种情况下，这些人这些社区相互交谈，这些社区相互交谈。所以它更像是一个关系网络。

01:20:08
That it is a geographical reality and a position.
这是一种地理现实和位置。

01:20:12
Right? So the idea of maps doesn't mean this. Now, that doesn't mean that they don't have a sense of geography, because all our records tell us that they will say.
对吧?所以地图的概念并不是这个意思。这并不意味着他们没有地理意识，因为我们所有的记录都告诉我们他们会说。

01:20:22
Okay, show us how to get from this place to this place. And then what you'll do is to go to, usually the travel by rivers, to go to the river bank, and you use a stick, and you draw up exactly.
好，告诉我们怎么从这个地方到那个地方。然后你要做的是，通常是沿河旅行，去河岸，你用一根棍子，准确地画出来。

01:20:33
Fox one. They tell you, add this tree, you make a left. You follow this river in this way until you hit this landmark, and then you make a turn.
狐狸。题目告诉你，加上这棵树，向左拐。你顺着这条河走，一直走到这个地标，然后拐个弯。

01:20:42
Can you make that landmark so they know how to navigate the train? It's just that you're not thinking about it in this in from a birth's ideal, right? Go west this much, and then go north, is this much? And go southeast, this much, right?
你能做个路标让他们知道怎么在火车上导航吗？只是你没有从出生的理想角度来考虑这个问题，对吧？向西走这么多，然后向北走这么多？东南方向，这么远，对吧?

01:20:55

What they're seeing is relationships this tree, and then that mountain, and then this nest that looks like that, and then this cave that looks like that.
它们看到的是关系这棵树，然后是那座山，然后是这个样子的鸟巢，然后是这个样子的洞穴。

01:21:05
So it's a series of relationships.
这是一系列的关系。

01:21:07
Right? So the way that we think about space will be very different based on culture, based on group, based on this, molecules, based on what people think is important.
对吧?所以我们思考空间的方式会有很大的不同基于文化，基于群体，基于分子，基于人们认为重要的东西。

01:21:18
All of these cultures, relationship is the most. Keeping Relationship builds who you are, defines who you are.
所有这些文化中，关系是最重要的。保持关系能塑造你，定义你是谁。

01:21:25
Right, not necessarily where you are, right? That makes you who you are, which is why, when Europeans came in and said, I belong to this land.
对，不一定是你所在的地方，对吧？这就是你是谁，这就是为什么，当欧洲人进来说，我属于这片土地。

01:21:35
And they're like, like, you define yourself by living in this space, why not by who you are related to?
他们就像，就像，你通过生活在这个空间来定义自己，为什么不通过你和谁有关系来定义自己呢?

01:21:43
Right and who you work with, right? Because for them, the web of

relationships is much more constituative of your identity.
对，还有你的同事，对吧？因为对他们来说，人际关系网更能构成你的身份。

01:21:53
And where you were born.
还有你出生的地方。

01:21:54
So the fascinating question, which means there are some maps in the world and many maps in the world that look like this that would never make sense of. GS.
这是一个有趣的问题，这意味着世界上有一些地图世界上有很多地图看起来像这样，这是无法理解的。GS。

01:22:07
Never, it just wasn't just.
从来没有，只是不只是。

01:22:12
Does he reflect?
他会反思吗?

01:22:12
Geographic, COVID. It's not matter, because that's just not how it's seen.
地理,COVID。它不是物质，因为它不是这样被看到的。

01:22:18
Is that fun?
有趣吗?

01:22:25
Very complex. That's why the whole module Office.
非常复杂。这就是为什么整个Office模块。

01:22:31
Okay, I'm going to, let's take a break here. And then we'll go do some exercises. We go play around some of the data in the software. Let's see.
好吧，我要，我们休息一下。然后我们会去做一些练习。我们在软件中使用一些数据。让我们看看。

01:22:43
5 min. We'll come back.
5分钟。我们会回来的。

01:22:49
Cities too.

01:22:50
But any project.
但是任何项目。

01:22:53
Now what we're going to do first, let me run you through the process of what we're going to do, but we're going to do this thing called Joe referencing.
现在我们首先要做的是，让我给你们演示一下我们要做的过程，我们要做的是Joe引用。

01:23:04
Right? Your referencing is taking a historical map and giving it coordinates, telling a computer.
对吧?你的参照是取一张历史地图，给它坐标，告诉电脑。

01:23:12
Essentially how to map, how to map, how to warp a map so that it fits the globe, right? The actual globe, right?
本质上是如何映射，如何映射，如何扭曲地图使其适合地球，对吧？真正的地球仪，对吧?

01:23:21
Telling it, where it, what cases correspond to what?
告诉它，在哪里，什么情况对应什么?

01:23:26
Let me fine.
让我好好想想。

01:23:38
So this is the map that we're going to work with today.
这就是我们今天要研究的地图。

01:23:46
This is a historical map from 1845, right.
这是1845年的历史地图。

01:23:52
So what we're going to do is we are going to try and figure out, you can see, there's a lot of information here, right, like the village peers that are abandoned.
所以我们要做的是，我们要试着弄清楚，你可以看到，这里有很多信息，对吧，比如被遗弃的村庄同伴。

01:24:02
Roles and stuff, right? So there's historical information here, which can be quite interesting, right? To see what's changed in the years.
角色之类的，对吧？这里有一些历史信息，很有趣，对吧？看看这些年发生了什么变化。

01:24:11
Since. Okay, so we're going to take this map and we're going to map it onto a war map.
自。好了，我们要把这张地图映射到战争地图上。

01:24:18

So what we're going to do is we're going to go here to huge is we're going to go to plugins.
所以我们要做的是我们要去这里，我们要去巨大的，我们要去插件。

01:24:25
I'm going to install plug in.
我要安装插件。

01:24:28
Okay. So plugins management.
好吧。所以插件管理。

01:24:34
All right.
好吧。

01:24:34
Open a menu. We're going to look for something that's called Quick Map Services. Curious.
打开菜单。我们要找一个叫做Quick Map Services的东西。好奇。

01:24:40
So quick.
那么快。

01:24:43
Q, U, I, C, K, M, A, P.
Q, u, i, c, k, m, a, p

01:24:51
Give me this one.
给我这个。

01:25:02
Hollow. If you need help, I guess I will keep proceeding. So once you're fun,

quick.
空洞。如果你需要帮助，我想我会继续。所以一旦你觉得有趣了，就快。

01:25:16
Have to.
必须的。

01:25:18
Yes, it's this one. Okay. Person. Stop.
是的，就是这个。好吧。的人。停止。

01:25:29
So once you find it, present, store. Okay, okay, you can install the plug in. It take a little bit of time, but not too much. It's not very big.
所以一旦你找到了，就送去，存起来。好了，好了，你可以把插头装进去了。这需要一点时间，但不会太多。它不是很大。

01:25:38
Okay.
好吧。

01:25:54
You may want to change the zoom of your.
您可能想要更改的缩放。

01:25:58
She can't find the author at the bottom.
她在底部找不到作者。

01:26:04
Anyone having issues installing the plugin?
有人安装插件有问题吗?

01:26:15
Very good, yes, but that.

很好，但是。

01:26:32
So while people figure out the plug in, does anyone know what a plug in is?
大家都知道插件是什么，有人知道插件是什么吗?

01:26:40
What's a plug in? It's not a library that you can download.
什么是插件？它不是一个你可以下载的库。

01:26:46
So a plug in is essentially an external tool that someone else has designed, right? That will help you do things very quickly, right? So it's not in the base functionality of.
所以插件本质上是别人设计的外部工具，对吧？这会帮助你快速完成任务，对吧？所以它不在基本功能中。

01:26:58
Futures itself. It's kind of an add on, right? It's a plug in that you can plug in so you can do things, convenient, things.
期货本身。这是一种附加，对吧？它是一个插头，你可以把它插进去，这样你就可以做一些事情，方便的事情。

01:27:07
In this case, quick Map Services is a very quick way of adding a map, of finding a map of policies. Okay, so once you've installed.
在这种情况下，quick Map Services是一种非常快速的添加地图、查找策略地图的方法。好的，一旦你安装好了。

01:27:15
Frequent services. You will be able to see it here.
频繁的服务。你可以在这里看到。

01:27:20
What if the equipment doesn't show up?

如果设备没出现怎么办?

01:27:25
Do not in your.
不要在你的。

01:27:39
Seeing the map itself. So you have to plug in, but you cannot find the map.
看到地图本身。所以你必须插入，但你找不到地图。

01:27:44
Oh, that's cute. Oh, that's not the same. This is for people who are in the field, who are adding data to.
哦，真可爱。哦，那不一样。这是为那些在这个领域工作的人准备的，他们正在向。

01:28:08
Okay, um.
好的,嗯。

01:28:21
All right, who is stuck in the plug, in? Plug, in, plug, in, plug, in, plug in? Oh, what is it with this? Okay.
好了，谁卡在插头里了？插，插，插，插，插，插？哦，这是怎么回事？好吧。

01:28:49
Okay, um.
好的,嗯。

01:28:49
So what we've done is we have loaded what we call base map, right? A base map is just.
我们所做的就是加载base map，对吧？底图只是。

01:28:57
One way of displaying the grid, right? So remember this model here.
一种显示网格的方式，对吧？记住这个模型。

01:29:01
You Jazz really loads this model. We have this model. It's just, it was blank. You didn't see anything because we have no projection.
你爵士真的加载了这个模型。我们有这个模型。只是，它是空白的。你什么也看不见，因为没有投影。

01:29:12
We have no visualization of the model, right? The grip exists. It's just invisible. And what we've done is we have put open street map, right?
我们没有模型的可视化，对吧？握力是存在的。它是看不见的。我们所做的就是放一张开放的街道地图，对吧?

01:29:21
Which is something that looks very much like Google Maps. So now this makes sense. You kind of know where we are on the coordinate grid.
它看起来很像谷歌Maps。这就说得通了。你大概知道我们在坐标网格上的位置。

01:29:29
How did you get this map? So once you can stop.
你怎么得到这张地图的？所以一旦你可以停下来。

01:29:35
Cream app sources up here.
奶油应用程序资源在这里。

01:29:37
The middle option, middle class Open Street Map. Stand up.
中间选项，中产阶级开放街道地图。站起来。

01:29:44

Okay, all right, I will move on. We can trouble shoot as we go along.
好吧，好吧，我继续。我们可以边走边解决问题。

01:29:53
What we're going to do is we're going to look for this place that the map is okay, what the husy, which is in Ontario, in Canada, I don't know.
我们要做的是找这个地方，地图上是这样的，在加拿大的安大略，我不知道。

01:30:04
I'll come a little bit.
我会来的。

01:30:06
Of American geography, right? Usually what I do if I'm not familiar, I just go on Google Maps and I quickly search Web Porter, who's here.
美国地理，对吧？通常情况下，如果我不熟悉，我会打开b谷歌地图，然后快速搜索Web Porter，看看谁在这里。

01:30:15
Right? You should be able to find it. It's what I suggest you do.
对吧?你应该能找到它。这是我建议你做的。

01:30:25
No big deal, where it is.
没什么大不了的，它在哪里。

01:30:26
You'll be joining yourself off. I used to plead your guess. So yes, it was just fun for a while.
你会自己离开的。我过去常常恳求你的猜测。是的，那只是一段时间的乐趣。

01:30:37
Okay, we will reach.

好的，我们会到达。

01:30:40
I know we don't like a controller.
我知道我们不喜欢控制器。

01:30:45
Oh, I can put the coordinates on something. Yes. So we have to close this looking for.
我可以把坐标放上去。是。所以我们要结束这个案子，寻找。

01:30:56
Okay. So if you put your cursor over a part of the map, I can do it because I make sure you guys. But here you see that that's a coordinate where your pointer is.
好吧。如果你把光标放在地图的一部分上，我可以这样做，因为我确保你们。但这里你看到的是指针所在的坐标。

01:31:11
At the bottom here. Okay.
在底部。好吧。

01:31:17
Oh, god, no, yes.
哦，天哪，不，是的。

01:31:27
All right, on the board. So you can see, as I hoggle.
好的，在黑板上。所以你可以看到，我在挣扎。

01:32:03
So.
所以。

01:32:03

Okay, that's the coordinates, right?
这就是坐标，对吧?

01:32:08
Where can we key in the coordinate?
我们可以在哪里输入坐标?

01:32:33
That doesn't work. It does. It's your current custom location. It doesn't actually zoom to your Oh, it doesn't zoom into it.
这行不通。它的功能。这是你当前的自定义位置。它不会放大到。哦，它不会放大到。

01:32:41
Let me try remember what it was okay, you know, you know, how I would do it. I would come here, and then I would go, huh?
让我试着回忆一下，你知道，你知道，我会怎么做。我来这里，然后离开，对吧?

01:32:51
Right? And then I'll look at that.
对吧?然后我来看看这个。

01:32:59
I'll be like, okay, where are we? Ah, okay. So it's here. Then I'll go back to QJS.
我会说，好吧，我们在哪儿？啊,好吧。它在这里。然后我会回到QJS。

01:33:09
And I'll see, okay, for all this here, right?
我看看，好吧，所有这些，对吧?

01:33:24
No, no, okay.
不，不，好吧。

01:33:26
You see this line here. This is the line between the US and Canada for now, right?
你看这条线。这是美国和加拿大之间的分界线，对吧？

01:33:36
It is to the left of this line.
它在这条线的左边。

01:33:38
I can see some. Okay, go out.
我能看到一些。好了，出去吧。

01:33:56
You probably added the map, really, because it's blue color. So Zoomer.
你可能添加了地图，因为它是蓝色的。所以放大镜。

01:34:04
It's more small, small, no, you're zooming in.
它更小，小，不，你放大了。

01:34:10
What do you know? I?
你知道什么？我吗？

01:34:13
I paint the number in this and it.
我把数字画在这里。

01:34:25
So we can do is we can.
所以我们能做的就是我们能做的。

01:34:36

Oh, so this is your skip group, right? So if you double click that, or you wreck, click for Zoom to layer, it will bring you the biggest extent of that layer.
这是你的跳跃小组，对吧？如果你双击它，或者点击缩放到图层，它会显示那个图层的最大范围。

01:34:49
Okay, so you can zoom in and find it. So be here.
好的，你可以放大并找到它。所以请留在这里。

01:34:54
Okay, okay. Thank you. All right. Help your friend.
好吧,好吧。谢谢你！好吧。帮助你的朋友。

01:35:04
Friends of stuff who hasn't found what those.
那些还没发现什么的朋友。

01:35:08
Okay.
好吧。

01:35:15
Okay.
好吧。

01:35:33
Title.
标题。

01:35:33
Very dangerous.
很危险的。

01:35:37

Oh, student, you go to two layers.
哦，同学们，你有两层。

01:35:56
Sure.
确定。

01:35:57
Oh.
哦。

01:35:58
Can I?
我可以吗?

01:36:03
So there are two parts of this. All we're going to do is we're going to load the map that we have.
这里有两个部分。我们要做的就是加载我们已有的地图。

01:36:16
Okay, so we're going to add a rester.
我们要添加一个休止符。

01:36:23
Because remember that.
因为记住这一点。

01:36:27
Remember that image files are resters.
请记住，图像文件是rest。

01:36:34
It's my.
这是我的。

01:36:38
Okay, I want you to add this one. Just add the GPA.
好的，我要你加上这个。只要加上GPA就行了。

01:36:49
Or the gypsies. I use the jyp like, that's fine. Okay.
或者吉普赛人。我使用jyp，这很好。好吧。

01:36:54
Okay, double click. That.
好的，双击。那。

01:36:58
Okay, so you have loaded the image. This is the map that you want to.
你已经加载了图像。这是你想要的地图。

01:37:07
Pure.
纯洁。

01:37:10
I only see one one board. I can't see this second board. Only one board.
我只看到一块黑板。我看不到第二块黑板。只有一块板。

01:37:25
No, no, true, um, christmas.
不，不，是真的，圣诞节。

01:37:30
Hello.
你好。

01:37:33
Oh, table. You need to open your table.

噢,表。你需要打开你的桌子。

01:37:42
Oh, okay. Thank you. Okay. Has everyone loaded your image?
哦,好的。谢谢你！好吧。每个人都加载你的图片了吗?

01:38:10
Okay.
好吧。

01:38:10
So now you've learned your image. What you're going to do is you're going to tell the computer which part of this map corresponds to, which part.
现在你已经学会了你的形象。你要做的就是告诉电脑地图的哪一部分对应哪一部分。

01:38:19
On the map itself. So what that means is I am going to select a point here.
在地图上。这意味着我要在这里选择一个点。

01:38:28
Let's say the top of this pier.
比方说这个码头的顶部。

01:38:31
Okay. And I'm going to go and select from map canvas. I'm going to tell it where on the actual map it is.
好吧。然后选择从mapcanvas。我要告诉它它在地图上的位置。

01:38:40
That press. Okay, so there's a written point here.
这新闻。好的，这里有一个书面的点。

01:38:47
You see, what I've done, what I've done is I have said, there is a point on

this map here, let's say, and click here.
你看，我所做的，我所做的是，我说，在地图上有一个点，我们说，点击这里。

01:38:59
I'm going to go onto my map canvas, and I'm going to tell the computer that that point correspond to this conference.
我要在我的地图画布上，告诉电脑那个点对应这个会议。

01:39:05
To copy out the coordinate.
把坐标抄出来。

01:39:09
So we're matching, you're going to select many points on this map here, and we're going to tell visually, visually.
我们要匹配，你要在地图上选择很多点，我们要直观地判断。

01:39:22
Where it is on the actual world.
它在现实世界中的位置。

01:39:25
Okay, can you try again?
好的，你能再试一次吗?

01:39:31
I don't know what you're using.
我不知道你在用什么。

01:39:38
Don't be at the very possible.
不要急于求成。

01:39:42

Again, it depends on how, how.
同样，这取决于如何。

01:39:51
I have no idea.
我不知道。

01:39:52
So.
所以。

01:39:52
So maybe I'll pick this point here, which is where the rope crosses over into the harbour. I'll pick that.
所以我可以选择这个点，也就是绳子穿过港口的地方。我选这个。

01:40:05
Okay, and I'll go onto the canvas and see where is this road?
好的，我在画布上看看这条路在哪里?

01:40:12
Are you like guessing? Or you're using, like, observation? I'm observing.
你喜欢猜吗？还是你在用观察力？我观察。

01:40:25
I will.
我会的。

01:40:28
Living right?
生活对吗?

01:40:29
You see, this is where the historian comes in to tell the computer which point corresponds to, what is your?

你看，这就是历史学家告诉计算机哪个点对应的地方，你的点是什么？

01:40:40
Education as a story, usually a good 01:15 to 20.
教育就像一个故事，通常好01:15到20。

01:40:49
Depends on the scale. On the map that I did, I had about 600 points. I manually.
取决于规模。在我绘制的地图上，我有大约600个点。我手动。

01:40:58
Back 600 points, but it was a continental map.
后退600点，但这是一张大陆地图。

01:41:02
And it was important that we thought it right.
我们认为正确是很重要的。

01:41:05
Yes. So you know this, I think three points, right? The points are here. So what we're telling the computer is.
是。你知道这个，我想有三点，对吧？要点在这里。所以我们告诉计算机的是。

01:41:17
This point is definitely there on the map. This point is definitely there on the map. And then all the points in the middle that we didn't tell them where it was, the computer will try and guess.
这个点在地图上肯定有。这个点在地图上肯定有。然后中间所有我们没有告诉它们的点，计算机会试着猜测。

01:41:26
And how it fitted in between them so that my script isn't part of it, it might stretch in part of it to conform to the points that you have to them. This is

the truth.
以及它是如何在两者之间协调的，所以我的剧本不是其中的一部分，它可能会在其中的一部分延伸，以符合你对它们的要求。这是事实。

01:41:37
This is the ground truth. So we're trying to what the map on top of the reality.
这是基本事实。所以我们试着在现实的基础上绘制地图。

01:41:47
Right? Because if you want to pull information from this historical map, you need to get a sense of where it will be treated in today's reality.
对吧?因为如果你想从这张历史地图中提取信息，你需要知道它在今天的现实中会被怎样对待。

01:41:56
Makes sense philosophically.
哲学上讲得通。

01:42:02
No.
否。

01:42:04
That's exactly what we're doing. We're super imposing, but you must super impose accurately, right? The question is, how?
这正是我们正在做的。我们是超级霸道，但你必须超级精准，对吧？问题是，怎么做?

01:42:16
How do you do the accuracy? Do the accuracy by pointing out to the computer where the points are the same to the right? so they're putting a page right through the map of the Earth.
你怎么做到准确？通过向计算机指出右边的点是相同的来确定精度吗？所以他们在地球地图上翻了一页。

01:42:28
And saying, this is what I have communicated to be true. This corresponds to this. This corresponds to this. This corresponds to this. If you do across the whole map, 15 to 20.
然后说，这就是我所说的事实。这个对应这个。这个对应这个。这个对应这个。如果你在整个地图上做，15到20个。

01:42:38
You probably get a decent approximation. It's a good question.
你可能得到一个不错的近似值。这是个好问题。

01:42:51
Because if the structures are different, then you have to judicate Are these exactly the same structures? There are a lighthouse there, and there's a lighthouse here today. Is it the same lighthouse? Can I use a coastline?
因为如果结构不同，你就必须判断它们的结构是否完全相同？那里有一座灯塔，今天这里也有一座灯塔。是同一座灯塔吗？我可以使用海岸线吗?

01:43:01
Oh.
哦。

01:43:09
What about the junction? Right? What do I think that's my problem.
交接处呢？对吧?我觉得这是我的问题。

01:43:16
Yes, I did. I had principles that I worked on. So I said, I'm going to pick these. This is my rationale, right? This category of this.
是的，我做了。我有自己的原则。所以我说，我要选这些。这是我的基本原理，对吧？这一类的。

01:43:31
And this is my reasoning, why I think they haven't shifted.

这就是我的理由，为什么我认为他们没有改变。

01:43:37
What are they?
它们是什么?

01:44:21
Decisions. You need to know enough about the situation.
决策。你需要对情况有足够的了解。

01:44:31
Well, well.
嗯,很好。

01:44:39
Okay. I only can see the point. I don't know how you start. Okay. You need to press at the point.
好吧。我只能看到这一点。我不知道你怎么开始的。好吧。你需要按一下这个点。

01:44:52
No, that's the lead you see. Press this one on the left. Click that. Okay. Click our point.
不，那是你看到的线索。按左边这个。单击。好吧。点击我们的点。

01:45:01
What happened here?
这里发生了什么?

01:45:04
But you don't have a point here.
但你说的没有道理。

01:45:13
Okay, um.

好的,嗯。

01:45:25
When you close out of this and reopen with your reports again, close it.
当您关闭此窗口并重新打开报表时，请关闭它。

01:45:30
Oh, because you're okay, okay, I get Press cancer. Press cancer. And then open the jar often again.
因为你没事，我得了媒体癌。按癌症。然后经常打开罐子。

01:45:38
In this Okay.
在这个好的。

01:45:43
Look the image, the rest image.
看看这个图像，剩下的图像。

01:45:46
So at the moment.
所以现在。

01:45:50
Okay, so now pick a point that you want to.
好，现在选一个点。

01:46:02
Exactly, okay, and then select on the map, move the window away. I think it's because it's.
没错，然后在地图上选择，把窗口移开。我想这是因为。

01:46:12
Okay, find the JQ dress window.
好的，找到JQ的礼服窗口。

01:46:16
Okay, then find that point on the map. We need scrolling for the.
好吧，那就在地图上找到那个点。我们需要滚动。

01:46:24
So what you've done is you click the point, and you've told it, that is that point, but it's not. Oh, so click that again.
你所做的就是点击这个点，然后告诉它，这就是那个点，但它不是。再点一下。

01:46:33
Okay, go back to here.
好的，回到这里。

01:46:35
Okay, fine. Where it is. So it's not clear.
好的,很好。它在哪里。所以还不清楚。

01:46:48
Oh, it's here.
哦，在这儿。

01:46:51
It's here. So what you want to do is you want to go back to QGIS. You want to go back to your job concern.
这里的。所以你要做的就是回到QGIS。你想回到你的工作上。

01:47:08
This one. Okay, yes. So try that again, because you closed out of it. So click a point that you want to do. Reference.
这一个。好吧,是的。所以再试一次，因为你关闭了它。点击你想做的点。参考。

01:47:17

It is. And I clear that, and then find that point here.
它是。我把它清除，然后找到这个点。

01:47:22
You need to Oh, you click something. Oh. So if I found that point, it can.
你需要点什么。哦。如果我找到了这个点，它就可以。

01:47:33
So, okay. So let me, let me do it once for you. Okay, I'll click that. Okay.
所以,好的。让我，让我为你做一次。好的，点击它。好吧。

01:47:43
I'm going to come here. I think it's around here. See, so why? When I click that, that coordinate comes here, and I'm going to press. Okay.
我要到这里来。我想就在这附近。看，为什么？当我点那个时，坐标会到这里，我要按。好吧。

01:47:52
Oh, okay. Thank you. Okay. Anyone having issues with any points? Are we good?
哦,好的。谢谢你！好吧。有人对要点有意见吗？我们还好吗?

01:47:59
Oh.
哦。

01:48:13
Oh, that's just to tell you, your problem is to have this comfortable.
哦，我只是想告诉你，你的问题是要过得舒服点。

01:48:40
It just depends, kind of, where it always to what the actual.
这取决于，它的实际位置。

01:48:45

So don't worry about don't worry about the red lines. You see the red lines. That's fine. It's meant to be okay.
所以不要担心不要担心红线。你看这些红线。这很好。这意味着一切都会好起来。

01:48:57
Okay.
好吧。

01:48:57
Oh.
哦。

01:49:21
Okay, I'll try to do about ten points. It's pretty addictive, actually. It's not like.
好吧，我尽量做到10分左右。实际上，它很容易上瘾。这不是。

01:49:43
Differences and such, right? What else?
差异之类的，对吧？还有什么？

01:49:46
So I want to do it.
所以我想这么做。

01:50:06
I.
我。

01:50:13
You know what?
你知道吗？

01:50:14

Okay, so.
好的,所以。

01:50:24
Everyone's concentrate on finding your points. We'll give it one more minute, because we definitely want to move on to the next one.
每个人都专注于找到自己的观点。我们再给它一分钟，因为我们肯定想继续下一个。

01:50:34
There is usually you want to rotate it before you put it in. Doesn't matter, because your ground, your points, will orient it the right way.
通常你想在放入之前旋转它。没关系，因为你的立场，你的观点，会指引正确的方向。

01:50:54
I don't think so.
我不这么想。

01:51:03
I don't think there is okay anyone having issues with finance with putting points down.
我不认为有人在财务上有问题而把分数记下来是可以的。

01:51:15
All right, okay. So what we're going to do, right? So let's recap what we've done. Okay. What we've done is we've created 1015 points.
好吧，好吧。所以我们要做什么，对吧？让我们回顾一下我们所做的。好吧。我们创造了1015个点。

01:51:27
Where we tell the computer that this point on the map corresponds to this point on the Earth.
我们告诉计算机地图上的这个点对应地球上的这个点。

01:51:33
Okay, there are questions around how you decide which points to use, right? Like, do I use a road? Do I use a junction? Do I use a tree?
好吧，关于如何决定使用哪个点还有一些问题，对吧？比如，我要走路吗？我要用连接点吗？我要用树吗？

01:51:44
Do I use a lighthouse, right? Do I use the coast or the river, right? And then there are questions around Is that accurate or not? Because a river may erode a coastline may change.
我要用灯塔吗？我应该走海岸还是河流，对吧？然后就有问题了这到底准确不准确？因为河流可能会侵蚀海岸线可能会改变。

01:51:56
Right, a building might get demolished. So it's up to you, as a historian, to exercise your judgment as to what you think are good points to use.
没错，一栋建筑可能会被拆除。所以，作为一名历史学家，你需要运用自己的判断力，判断哪些观点值得使用。

01:52:04
Okay. So what we've done is we have found ten points of correspondence. And what we're going to do is we're going to tell the map to do the rest right. We are going to now get bit to work.
好吧。我们找到了十个对应点。我们要做的是告诉地图把剩下的部分做好。我们现在开始工作。

01:52:14
The map. So what we're going to do now is we're going to go here.
地图。所以我们现在要做的是我们要去这里。

01:52:17
Two settings, and then we're going to go into transformation settings.
两个设置，然后我们会进入变换设置。

01:52:26

Right? You're going to pop up this window here, and this will actually tell the computer how you wanted to do this transformation work, right? I want you to take polynomial to.
对吧?你会弹出这个窗口，它会告诉电脑你想要怎样做这个变换，对吧？我想让你们取一个多项式。

01:52:37
Right? This is the only thing you need change transformation type polynomial tools. These are just different algorithms, right? Some algorithms are better for certain maps, so some maps that are much Bigger.
对吧?这是唯一需要改变变换类型的多项式工具。这些只是不同的算法，对吧？有些算法对某些地图更好，有些地图更大。

01:52:51
You use something like ten, six points, you don't need to know the details, right? Just don't do linear. Linear is probably not great for this. Historical maps is not great for maps that are more modern, linear is okay.
你可以用10点，6点，你不需要知道细节，对吧？只是不要做线性。线性可能不是很好。历史地图并不适合现代地图，线性地图还可以。

01:53:02
But historical maps, generally, you won at least want the polynomials, usually two or three. Okay, so I'm going to just pick two for now.
但是历史上的地图，一般来说，你至少需要多项式，通常是两个或三个。好的，我现在只选两个。

01:53:09
I'm going to press. Okay. So I've saved the settings. Okay.
我要按了。好吧。我保存了设置。好吧。

01:53:16
You see, the lines have changed. And then what I'm going to do is I'm going to press play.
你看，台词变了。然后我要做的是按播放。

01:53:21
Stop your I can see.
停下你的我能看见。

01:53:24
And what you notice is, if you go back here.
你会注意到，如果你回到这里。

01:53:30
Oh.
哦。

01:53:36
Yours is probably better than mine, because I did a shit job at this. But what we're going to do is really go to properties.
你的可能比我的好，因为我做得很糟糕。但我们要做的是进入属性。

01:53:47
Go to symbol. Go to legend. Could I add this thing called opacity slider?
转到符号。进入传奇。我能添加这个不透明度滑动条吗?

01:53:56
Okay. What this does allows me to quickly.
好吧。这能让我很快。

01:54:03
Check. Oh.
检查。哦。

01:54:07
So to get opacities, whether you write, click on the image itself, right, is that your reference image good property.
要获得不透明度，无论你写什么，点击图片本身，对吧，这是你参考图片的好属性。

01:54:20
With a legend, and then.
有了传说，然后。

01:54:25
Better take the right here so you can add it, and I've got the.
最好从右边开始，这样你就可以把它加进去，我得到了。

01:54:33
Oh.
哦。

01:54:34
Okay, and you can see how you've done.
好了，你可以看到你是怎么做的。

01:54:38
How did you guys do right? Click the your reference image.
你们是怎么做到的？单击您的参考图像。

01:54:56
For the properties, you can probably find $2 with it.
对于属性，您可能可以找到2美元。

01:55:05
All right, but this is essentially it. That's the whole PET line. I've got a map. I set my.
好的，但本质上是这样的。这是整个PET生产线。我有一张地图。我设置了我的。

01:55:17
Gcps, the rip dots, right? The control points. These are points of truth, the ground truth.
Gcps，撕裂点，对吧？控制点。这些都是真理，基本真理。

01:55:24
And then I choose how I want the computer to do the rest of it, walk the rest of it. And then I first play, and then you will do the walking.
然后我选择如何让电脑完成剩下的部分，走完剩下的部分。然后我先玩，然后你走。

01:55:34
And now I've done the washing. I can go back to the map, right? Of course, I double check and see what I should do a better job. In this case, I probably should not, right? So I can do it. If I can do that, you're concerned.
现在我已经洗完衣服了。我可以回到地图上，对吧？当然，我会仔细检查，看看我应该在哪些方面做得更好。在这种情况下，我可能不应该这样做，对吧？所以我可以做。如果我能做到，你就会担心。

01:55:45
So if you feel that this is not ideal.
如果你觉得这不理想。

01:56:12
You can go back. You can add more points. If you think that a point was done wrong. You can edit the point. You can shift the points right.
你可以回去。你可以再加点。如果你认为某一点做错了。你可以编辑这个点。你可以向右移动这些点。

01:56:19
That's all the Europe.
这就是整个欧洲。

01:56:20
Range of time.
时间范围。

01:56:24
We shall finish with your referencing here, but you get the process right.

We have a map. We tell the computer which points correspond to what?
我们将在这里结束你的引用，但你的过程是正确的。我们有一张地图。我们告诉计算机哪些点对应于什么?

01:56:33
On the Earth and the GS, and then we tell the computer to run an algorithm to do the walking. Okay?
在地球和GS上，然后我们告诉计算机运行一个算法来进行行走。好吧?

01:56:42
So what it means is now I have this historical map that is kind of accurate, right? I can pull out the details from this map now. So if, let's say, there is a point here that says, there's a lighthouse here.
这意味着我现在有了这张历史地图这张地图比较准确，对吧？我现在可以从地图上找出细节了。假设这里有一个点，这里有一个灯塔。

01:56:54
I can actually plot the lighthouse where it will be today.
我可以画出灯塔现在的位置。

01:56:58
Realize that, and then you bring it back.
意识到这一点，然后把它带回来。

01:57:45
Ah, there we go. Very good. Thank you. So minimize the future. And I open that, and then this dialogue box.
啊，好了。很好。谢谢你！所以最小化未来。我打开这个，然后是这个对话框。

01:57:57
Okay, good. Okay. So if I wanted to shift a point, right, I can go to here. I've been adding.
好的,很好。好吧。如果我想移动一个点，可以到这里。我一直在加。

01:58:07
If I look here, I can shift. So I say, maybe this point was not great. So I'm going to shift this point.
如果我看这里，我可以移动。所以我说，也许这个点不是很好。我要平移这个点。

01:58:13
Go to the point, hold it down, and move it. That's why I say a mouse.
找到那个点，按住它，移动它。这就是为什么我说老鼠。

01:58:25
Oh.
哦。

01:58:29
We will update the rule coming. The rule is just a representation of what's here. This is the actual data. You're just doing it properly.
我们将更新规则。这个规则只是这里的一个表示。这是实际数据。你做得很好。

01:58:41
Hold it and move it.
抓住它，移动它。

01:58:49
Okay, you can also do it on that map. If you need to change the one, you can also delete points. So the middle option here is to delete a point.
你也可以在地图上做。如果需要更改一个，还可以删除点。中间的选择是删除一个点。

01:59:01
If you think that point was terribly done, take that again.
如果你认为这一点做得很糟糕，再做一次。

01:59:06

Okay. So something that's interesting to note is that you've done this work, right? Creating all these points. You can see the points that said, data about.
好吧。有趣的是你已经做过这个工作了，对吧？创建所有这些点。你可以看到这些点，数据关于。

01:59:18
Okay, as a CSB, and then you can transfer it somewhere else. So as long as you have the GCP, you can just quickly load the GCP file to QJ.
好吧，作为CSB，然后你可以转移到其他地方。因此，只要您有GCP，您就可以快速将GCP文件加载到QJ。

01:59:30
And tell them to do it over again. GCP Ground Control point. These are the right points that you've been doing.
让他们再来一遍。地面控制中心。这些是你一直在做的正确的点。

01:59:39
The truth, the ground truth.
真相，地面的真相。

01:59:42
So that's the list of points that you have ascertained are true correspondences.
这就是你确定的对应点的列表。

01:59:50
This point means that on the world, that point is that on the world, you can save that as a separate part of the GCP part. It's just a text file as a CFP.
这一点意味着在这个世界上，这一点是在这个世界上，你可以把它保存为GCP部分的一个单独的部分。它只是一个作为CFP的文本文件。

02:00:01
Okay.
好吧。

02:00:01
So that's your referencing fun, promising. Okay.
这就是你的引用有趣，有希望。好吧。

02:00:10
So you can see it, see, and then I see why you see it as well. They call it a GCP file like it's just, you can open it in any future as well.
所以你可以看到它，然后我明白为什么你也能看到它。他们称它为GCP文件，你可以在任何时候打开它。

02:00:23
Okay, so this is your referencing. All right, you can go home and play with it. But what we want to do now is we're going to have a look at one last thing.
好了，这就是你的引用。好了，你可以回家玩了。但我们现在要做的是看最后一件事。

02:00:36
All right, we might, I don't know whether we would just do this as scary, but okay, so.
好吧，我们可能会，我不知道我们会不会因为害怕而这么做，但是好吧，那么。

02:00:59
I'll give you another photo. It's called CN Maps, or something like that. This is an actual map that dr Yip was trying to recreate for her monograph. So she's ever published a book.
我再给你一张照片。叫做CN地图之类的。这是叶博士试图在她的专著中重建的一幅真实的地图。她曾经出过书。

02:01:10
She has this map here. It's called the Tourist Map of Changjiang River. And she says, look, I need to put a map in my book. This has some of the data points that I want. I don't need all of these different things. I only need

some of these.
她有这张地图。它叫《长江旅游地图》。她说，看，我需要在我的书里放一张地图。这里有一些我想要的数据点。我不需要这些不同的东西。我只需要一些。

02:01:23
Right?
对吧?

02:01:24
And so the question is, how do we do this right? What she said was, I need this river.
所以问题是，我们怎么做才对？她说，我需要这条河。

02:01:34
I need this week. I need some of the cities like Shanghai, nanjing, wuhan.
我需要这个星期。我需要一些像上海、南京、武汉这样的城市。

02:01:43
The major cities that let painting.
各大城市都让绘画。

02:01:46
Okay. So very easily think of the data that you need. You need point data for the cities. You need line data.
好吧。所以很容易想到你需要的数据。你需要城市的点数据。你需要线数据。

02:01:56
For the rivers, right? And that's essentially it, right? It was just interesting, right?
为了河流，对吧？本质上就是这样，对吧？只是很有趣，对吧?

02:02:05
Sometimes it's a line because you don't close it. Sometimes it's a polygon

because you can put lines and polygons in the same data form. They are different.
有时它是一条线，因为你没有关闭它。有时它是一个多边形因为你可以把直线和多边形放在同一个数据形式中。它们是不同的。

02:02:14
Joe, jason, you can shape off. They're different. So if you open up the CN map.
乔，杰森，你们可以走了。它们是不同的。如果你打开CN地图。

02:02:25
The Internet, right? Which is why I asked you guys to have a look at.
互联网，对吧？所以我才让你们来看看。

02:02:32
Business.
业务。

02:02:32
So let's say, reopen.
我们说，重新打开。

02:02:35
Okay. So here we have points.
好吧。这里有点。

02:02:53
That's the line, the other set.
这是另一条线。

02:02:57
This is a ruster data set. You see the squares. That's a ruster.
这是一个生锈的数据集。你看这些方块。这是一个锈迹。

02:03:04

Okay, we don't actually have a polygon.
好的，我们实际上没有多边形。

02:03:07
Because none of this is called as a polygon. Okay, I'm going to hide the OSM map. Now it doesn't look so crowded.
因为这些都不是多边形。好吧，我要把OSM地图藏起来。现在看起来不那么拥挤了。

02:03:17
These all the points.
这些都是点。

02:03:23
Close that.
关闭。

02:03:23
These are the points that we potentially create.
这些是我们可能创造的点。

02:03:27
So.
所以。

02:03:27
These are the reverse. The major reverse. Just lines.
这些是相反的。大逆转。就行。

02:03:37
Is this also manually drawn. So each river lines.
这也是手工画的吗？所以每条河的线。

02:03:47
Data sets are available made by other people.

数据集是由其他人提供的。

02:03:50
Um, some of them, not necessarily some of them. You can use satellite imagery, and then you can do what we did GCP, and then it will map on that.
有一些，不一定有一些。你可以使用卫星图像，然后你可以做我们做的GCP，然后它会在上面绘制地图。

02:04:04
So that is possible. Some people work, E.G., vanessa loaded a QJS version called q few.
所以这是可能的。有些人工作，例如，vanessa加载了一个叫做qfew的QJS版本。

02:04:13
Right on her tablet. And what people will do will they will carry the tablet, and they'll go to that location, and they'll say, this is the location satellite. Tell me what the coordinates are right now.
就在她的平板电脑上。人们会做的是他们会带着平板电脑，他们会去那个位置，他们会说，这是定位卫星。告诉我现在的坐标。

02:04:25
So there's data capture in the view, right? You might go to an archaeological site, and you will say, this is a war that starts from here and then there. So I'm going to come here. I'm going to collect this point here.
视图中有数据捕获，对吧？你可能会去一个考古遗址，你会说，这是一场从这里开始的战争。我要到这里来。我要在这里收集这个点。

02:04:38
And head is a green of the wall. I'm going to walk five steps. I'm going to plot another point, or walk my steps. and caught another point that did I capture in a few.
而头是一堵绿色的墙。我要走五步。我要画出另一个点，或者走我的路。抓住了我在几分钟内抓住的另一个要点。

02:04:46
Right? We don't usually do that because we were taxed. We're archival materials.
对吧?我们通常不这么做，因为我们被征税了。我们是档案材料。

02:04:52
You can go and do some of this mapping yourself if you want to. Right? Geographers do it. Archaeologists do it because they're working with stuff in the real world.
如果你想的话，你可以自己做一些映射。对吧?地理学家就是这样做的。考古学家这样做是因为他们在研究现实世界中的东西。

02:05:03
Not like we don't, we don't work with the real world. Okay? But urban standards do it job. It does do it. People who work in policy, right? Our government meet this kind of data. This is a hydography data, right?
不是说我们不，我们不跟现实世界打交道。好吧?但城市标准起到了作用。它确实做到了。从事政策工作的人，对吧？我们的政府面对这样的数据。这是水文数据，对吧?

02:05:17
So where the waterways are in China in this case? Okay. So essentially, you're going to look at the time. We're not going to necessarily go through it.
那么，在这种情况下，中国的水道在哪里？好吧。所以本质上，你要看时间。我们不一定要讲一遍。

02:05:27
But what we do is we identify for Doppy. She wanted a map like this with the rivers and some of the major cities. Okay.
但我们要做的是为多普比识别身份。她想要一张这样的地图，上面有河流和一些主要城市。好吧。

02:05:36

The data sets, so I work with her on this project, are available online. They're very common, right? Cities and rivers, very common geographical data.
我和她一起做这个项目的数据集可以在网上找到。它们很常见，对吧？城市和河流，非常常见的地理数据。

02:05:49
So I don't have to go create it myself. I can go find the data set that someone else has created, a big one, and I can isolate the stuff that I Need.
所以我不需要自己创建它。我可以去找别人创建的数据集，一个大的数据集，我可以把我需要的东西分离出来。

02:05:58
Right? So what I'm going to do is I'm going to go into the data set. This is the big data set, right? I'm going to open it up. I'm going to open the attribute table.
对吧?所以我要做的是进入数据集。这是一个大数据集，对吧？我要把它打开。我要打开属性表。

02:06:09
An attribute table tells me all the different lines and all the rivers exist. So I can search this E.G.?
一个属性表告诉我所有不同的线和所有存在的河流。所以我可以搜索这个E.G.?

02:06:19
Right, and find a specific river.
对，找到一条特定的河。

02:06:22
So if I don't know where this river is.
所以如果我不知道这条河在哪里。

02:06:27
So here, this is the attribute. This is all the lines of data that exist.

这里，这是属性。这是所有存在的数据行。

02:06:47
You can edit the data. It's possible to edit the data.
您可以编辑数据。可以编辑数据。

02:06:55
You see the.
你看。

02:07:00
In this case, this is a very It doesn't have that much information. It's just geographical information. It doesn't mean.
在这种情况下，这是一个非常它没有那么多的信息。它只是地理信息。这并不意味着。

02:07:14
So this data set is fine for certain things, but it's not going to give you much more than just me.
所以这个数据集对某些事情来说很好，但它不能给你比我更多的东西。

02:07:20
If we look at let's see the cities, maybe let's see what we get.
如果我们看一下让我们看看城市，也许让我们看看我们得到了什么。

02:07:24
If we look at this one.
如果我们看这个。

02:07:26
So this is another data set of reverse, right? But just major reverse. I give you the name.
这是另一个反向的数据集，对吧？但只是大逆转。我给你名字。

02:07:37

But it also gives you the length of the river.
但它也给出了河流的长度。

02:07:40
These things are necessarily pop up on the map, right? They're not needed for putting it on the GS map itself. But this is additional data.
这些东西必然会出现在地图上，对吧？把它放到GS地图上并不需要它们。但这是额外的数据。

02:07:50
Right? That's associated with this river, in this case.
对吧?在这种情况下，它与这条河有关。

02:07:53
It's not data about this data set, so it's not meta data. They're just additional attributes, right? The reason why these attributes exist.
它不是关于这个数据集的数据，所以不是元数据。它们只是附加属性，对吧？这些属性存在的原因。

02:08:09
If you notice, the name is also an attribute. It's because when you're labeling something on a map, you can tell the computer what you want to choose as your label.
如果您注意到，这个名称也是一个属性。这是因为当你在地图上标记某物时，你可以告诉计算机你想选择什么作为你的标签。

02:08:18
Basically. So let's say this one, right? This one is the label, the name.
基本上。我们说这个，对吧？这个是标签，是名字。

02:08:29
Right?
对吧?

02:08:30

Okay. So you see, right now, this is just a line, right? What I can do is I can change the symbology.
好吧。你看，现在，这只是一条直线，对吧？我能做的就是改变符号。

02:08:45
At a legal.
在法律上。

02:08:49
So I've added a label for the name.
我给名字加了一个标签。

02:09:04
But what I can do is I can also change the label.
但我可以做的是，我也可以改变标签。

02:09:08
Change. What is shown is to the length and kilometers.
改变。所显示的是长度和公里。

02:09:14
So we can map, like, a lot of things. And if, probably, if you have different maps, you can see how, E.G., that's or like how it changed.
所以我们可以映射很多东西。如果你有不同的地图，你可以看到，比如说，它是如何变化的。

02:09:34
How then do you use this to tell a story? Because now, like, this is like, good for us, but, like, how would you convey this message to, like, a?
那么你如何用它来讲述一个故事呢？因为现在，这对我们来说是件好事，但是，你怎么把这个信息传达给，比如，a?

02:09:45
So let's say, let's say, here's a, here's a true line, right? We've got, let's say, a historical map of reverse in China from 500 years ago.

假设，假设，这是一条真线，对吧？比如说，我们有一张500年前的中国历史地图。

02:09:56
And then we do reference it, so we look at kind of where the cities are, and we know the cities haven't changed their location, right? So maybe we use cities as our gcps, where our control points.
然后我们参考它，所以我们看一下城市在哪里，我们知道城市没有改变他们的位置，对吧？所以也许我们可以用城市作为我们的gcp，我们的控制点。

02:10:09
And then from there, we are able to map Joe reference the map, so we can actually see where the historical rivers used to be, or the course that they used to take, and we know, within 500 years, if it's sort of.
然后从那里，我们可以画出乔参考的地图，所以我们可以看到历史上的河流曾经在哪里，或者它们曾经的河道，我们知道，在500年内，如果它有点。

02:10:21
Not if we don't interfere with the reverse with a change course. Right? So then we can compare it to a data set from today, which is this. And we can see how that?
如果我们不改变航向干扰它就不会。对吧?然后我们可以将它与今天的数据集进行比较，就是这个。我们可以看到这是怎么回事?

02:10:31
River has shifted or changed. And that could be interesting because, historically, if you have a source that says this person traveled.
这条河已经改变了。这可能很有趣，因为从历史上看，如果你有一个消息来源说这个人旅行过。

02:10:41
30 days to see this river. Where actually was the river, right? Is it this river that he was talking about? Or is it another river we never knew?
30天看完这条河这条河到底在哪里，对吧？他说的是这条河吗？还是另一条我们不知道的河?

02:10:50
Right? But was it possible? And by looking, taking information from a historical map and nothing, until today, we can actually do the analysis.
对吧?但这可能吗？通过观察，从历史地图中获取信息，直到今天，我们才能真正进行分析。

02:10:57
Let's say I want to show the change, like, maybe there's a change, right? Like, how would I communicate this to an?
假设我想显示变化，比如，可能有变化，对吧？比如，我该怎么跟他沟通？

02:11:05
You would, you would produce two images, probably one the historical one and one the new one, that you will overlay it. Do you have tools like that? Let's say you have multiple years.
你会，你会产生两个图像，可能一个是历史的一个是新的，你会把它们叠加在一起。你们有那样的工具吗？假设你有很多年。

02:11:18
So you have to turn on both layers. Yes, you can turn on both layers even more. And just suppose it.
这两层都要打开。是的，你可以把这两层都打开。假设一下。

02:11:24
So like that, e.g., let's say we've got this from a historical map, and we got this from another data set. You choose what information to present, and then you export that View.
就像这样，比如说，我们从历史地图上得到这个，我们从另一个数据集得到这个。您选择要显示的信息，然后导出该视图。

02:11:34
Right? And then that tells a story, right? So I could turn on this data set and show this one, and then this is one view, and I turn it off, and I put it in another data set that shows maybe this new speaker.

对吧?这就说明了一个问题，对吧？我可以打开这个数据集，显示这个，然后这是一个视图，然后我关闭它，我把它放在另一个数据集，显示这个新的扬声器。

02:11:46
And then that.
然后是这个。

02:11:48
As a change, and you will produce those two images, and you use that.
作为一种改变，你将产生这两幅图像，你使用它。

02:11:52
That's how you tell a story. The data is there to be visualized, and then you curate which data shows up.
这就是讲故事的方式。这些数据是用来可视化的，然后你来管理显示哪些数据。

02:12:02
How it shows up, what labels you put so on and so forth. We're going to visualization here, which is another bit. But even thinking about the labels.
它是如何显示的，你贴了什么标签等等。我们要在这里可视化，这是另一个比特。但即使考虑到标签。

02:12:11
Where the labels sit, does this label sit here, or does it at the end? That's a choice. Computer doesn't predicate you to. It's doing a particular thing, but you make that Decision.
标签在哪里，这个标签在这里，还是在最后？这是一种选择。电脑并不能预言你。它在做一件特定的事情，但是你做了决定。

02:12:22
Right?
对吧？

02:12:23
But the data does exist, so you can play around with I have a question about the attribute table. So this table is.
但数据确实存在，所以你可以试试我有一个关于属性表的问题。这张表是。

02:12:34
Produced ultimately, by this tool, right? Because, like we have different points in this we list them, and then it become an attribute table. Or.
最终由这个工具产生，对吧？因为这里有不同的点我们把它们列出来，然后它就变成了一个属性表。或。

02:12:44
Like we could import data set, like we already have some informations, for instance, like we have some.
比如我们可以导入数据集，比如我们已经有了一些信息，比如我们有一些。

02:12:53
Reverse information, and we can impulse in it, and it will show in the picture. Or we can, just like draw on this my.
反向信息，我们可以把它冲入其中，它会在图中显示出来。或者我们可以，就像画这个。

02:13:04
On this layer. So there are about three layers to this present. One is you.
在这一层。这个礼物大概有三层。一个是你。

02:13:12
Can draw yourself right, looking at the map, you can say, oh, there used to be a canal from this to this. I know this from historical sources, but no one has made a data set for that.
你可以自己画，看着地图，你可以说，哦，以前这里和这里有一条运河。我从历史资料中知道这一点，但没有人为此做过数据集。

02:13:22
So what I can do is I can start plotting a point like that. Follows this line

here, and it goes like that. And that's how you create the GS data that's creating. What I've done here, in this case is I've.
我能做的就是画出这样一个点。沿着这条线，像这样。这就是你创建GS数据的方式。在这个例子中，我所做的是。

02:13:32
Taken someone's data set, and I've just imported it. It comes with all these extra information and extra attributes. And what I can do is I can change how it looks by using the SYM Geographical Data.
取别人的数据集，然后导入。它带有所有这些额外的信息和额外的属性。我能做的就是通过使用SYM地理数据来改变它的样子。

02:13:43
That underpins it, but I can edit it. So, E.G., for dr Smith, right? That looks very constant, like this.
这是它的基础，但我可以编辑它。例如，史密斯博士，对吗？它看起来是恒定的，像这样。

02:13:52
I know she only wants this river, and she wants like four cities and a lake. So what I do is I go back to the big data set that I have, and I isolate exactly which ones I want, and I copy that out into.
我知道她只想要这条河，她想要四个城市和一个湖。所以我要做的就是回到我拥有的大数据集，把我想要的数据分离出来，然后复制到。

02:14:03
I filtered the asset, and then when you use that. So that's here. So let's turn this off, this.
我过滤了资源，然后当你使用它的时候。在这里。我们把这个关掉。

02:14:10
Oh, there's only one thought. Okay.
噢，我只有一个想法。好吧。

02:14:20

So this, these rivers come from the larger data set.
这些河流来自于更大的数据集。

02:14:32
Just taken them out and extracted them out.
把它们取出来，再取出来。

02:14:35
So I have this river here and Shanghai.
这里有这条河和上海。

02:14:39
City that I put out from another data set of cities, right? So I have two big data sets. They have too much information I put out the ones I need from the rivers. I pull out the ones I need from the city.
我从另一组城市数据中取出的城市，对吧？我有两个大数据集。他们有太多的信息，我把我需要的来自河流的。我从城里拿出我需要的人。

02:14:51
And then these are the two that I put up from the two that once.
然后这两个是我从这两个中抽出来的。

02:14:55
And then now I can use this.
现在我可以用这个了。

02:14:58
And you said, the cream, the map, does not need it.
你说，奶油，地图，不需要它。

02:15:04
Okay.
好吧。

02:15:04

Makes sense.
是有意义的。

02:15:08
All right. Any questions. Yes, I have a question. So how do you determine whether?
好吧。任何问题。是的，我有一个问题。那么，如何确定是否存在呢?

02:15:18
This data set is good question. The assets exist out in the world.
这个数据集是个好问题。资产存在于世界上。

02:15:26
You have to ask yourself, like any source who created it, why did they create it? What was the purpose behind it? You will find that GST assets tend to be created by hobby, yes, and people who love history and people who love geography.
你必须问自己，像任何创造它的人一样，他们为什么要创造它？这背后的目的是什么？你会发现，GST资产往往是由爱好创造的，是的，还有那些热爱历史和地理的人。

02:15:41
Which is great, because if you're looking for common stuff, it's probably authentic. The idea is that's for Singapore in Malaysia, because people love thinking about geography advice.
这很好，因为如果你在找普通的东西，它可能是真的。这是给马来西亚的新加坡的，因为人们喜欢考虑地理建议。

02:15:50
Right? You might ask yourself, is there a reason for them to lie about it or to be shabby about it? Right? If the ideas that are created by researchers, you probably Safe.
对吧?你可能会问自己，他们是否有理由对此撒谎或卑鄙？对吧?如果这些想法是由研究人员创造的，你可能会安全。

02:16:01
The assets, they are created by people who have done hundreds of the assets. You're probably safe If so, you have to do that. Gait yourself.
这些资产是由做过数百种资产的人创造的。你可能是安全的如果是，你必须这么做。步态自己。

02:16:10
It takes time to create these assets. So for something to exist, it needs to exist for a reason. Usually it's passion. Sometimes it's work. Sometimes it's research. Sometimes it's because governments make it.
创建这些资产需要时间。所以事物的存在，需要有其存在的理由。通常是激情。有时是工作。有时是研究。有时是因为政府做出了这样的决定。

02:16:21
And it's open data. The Singapore E.G. has open data. They have some GSDA sets, and not a lot, right? So they have GSDS, not, I don't know, temples in Singapore or Something.
这是开放数据。新加坡E.G.有开放数据。他们有一些GSDA，但不多，对吧?所以他们有GSDS，而不是新加坡的寺庙之类的。

02:16:31
Or hopefully this is useful.
希望这是有用的。

02:16:34
All right. Or heritage spaces, or master plan. Master plan is a chair. That's it. It tells you exactly which plot of land is, what and zone for what?
好吧。或者遗产空间，或者总体规划。总体规划是一把椅子。就是这样。它会告诉你哪块地是什么，有什么用途?

02:16:45
Right? So you have all these polygon Singapore is just split into different polygons, and each polygon has different attributes. So attribute purpose.
对吧?所有这些多边形新加坡被分成了不同的多边形，每个多边形都有不同的属性。属性目的。

02:16:54
Religion, side worship, or purpose education, or purpose housing, right, or purpose undesonated, so forth. Those are attributes.
宗教，宗教崇拜，或目的教育，或目的住房，权利，或目的不明，等等。这些都是属性。

02:17:05
They give you additional information about that polygon or that line, or that point, whatever.
题目给出了关于这个多边形，或者这条线，或者这一点的附加信息。

02:17:11
How do you all come across this? DNSA? So how do you find it?
你们是怎么想到这个的？DNSA吗?那么你怎么找到它呢?

02:17:18
I agree.
我同意。

02:17:23
Google, and then I looked at what I think will be useful, what I think will be accurate. So I had to adjudicate, in terms of.
b谷歌，然后我看我认为什么是有用的，什么是准确的。所以我必须做出裁决。

02:17:32
Whether this is good enough for what I need, right? Is it precise enough for what I need? Because what I need is essentially just a representation, especially of how things up in negative.
这是否足以满足我的需求，对吧？精确到能满足我的需求吗？因为我需要的本质上只是一个表示，特别是在负数的情况下。

02:17:43
She doesn't need to drive a boat down this river, so she doesn't need that

level of precision, even though this is actually quite precise, etc. If she wanted to, she could use it.
她不需要驾船顺流而下，所以她不需要那么精确，尽管这实际上是相当精确的，等等。如果她想的话，她可以用。

02:17:53
Okay, this part is sufficient.
好的，这部分是充分的。

02:18:05
So what you could do very quickly is you could use a basement like the OSM.
所以你可以很快做的是你可以使用一个地下室，比如OSM。

02:18:15
But you will notice that.
但是你会注意到的。

02:18:21
Some parts of it are not exactly the same.
有些部分是不完全一样的。

02:18:24
You see, actually, this is a very good data set, because the highlight to you where the islands are in the river, all the different parts of.
你看，实际上，这是一个非常好的数据集，因为突出显示的是岛屿在河中的位置，所有的不同部分。

02:18:32
River. It's a very detailed river data set.
河。这是一个非常详细的河流数据集。

02:18:36
See.
看到的。

02:18:36
But you will see that this is here.
但是你会看到这个在这里。

02:18:45
Right?
对吧?

02:18:48
Which one will you trust?
你会相信哪一个?

02:18:51
Do you trust the green one, which is a separate dataset that I quote from? Or we use trust the basement, which is the basement.
你相信绿色的那个吗，我引用的是一个独立的数据集？或者我们用“信任地下室”，也就是地下室。

02:19:02
Really.
真的。

02:19:03
So assuming the person who created the rivers the other side, the green one.
假设创造河流的人在另一边，绿色的那条。

02:19:15
Doesn't have any reason to lie. I will actually trust them more than the Open Street Map, because the Open Street Map is a global dataset. Remember, we were talking about the trade laws you have to make around accuracy.
没有理由撒谎。比起开放街道地图，我更相信它们，因为开放街道地图是一个全球数据集。记住，我们讨论的是关于准确性的贸易法则。

02:19:27
If something is geographically constrained, in this case only in China, then they are probably working.
如果某些东西受到地理上的限制，比如只在中国，那么它们可能是有效的。

02:19:35
On the assumption that you want to be precise within this area and not so precise outside, whereas open.
假设你想要在这个区域内精确而不是在外面那么精确，而开放。

02:19:43
Street map needs to be precise enough, everywhere.
街道地图需要足够精确，无处不在。

02:19:45
The basement is the one that's lying to you.
对你撒谎的是地下室。

02:19:49
And you use the China Medical I did. It's in a book. So we three days, we sat down, and I thought her GS from beginning to end. And then she spent one week.
你用的是我用的中医。它在一本书里。所以我们坐了三天，我从头到尾都在想她的GS。然后她花了一个星期。

02:20:07
Playing around and being addicted to doing gcps and making maps. And one year later, she's like, life got I said, they asked me to make some revision to this map. How do I do that?
到处玩，沉迷于做gcps和制作地图。一年后，她说，生活让我说，他们让我对这张地图做一些修改。我怎么做呢?

02:20:19
That's it. Okay, let's sit down. Very easy. And it's like, let me write. You

understand the principle, right? This is how you draw references. How is how you pull data from one copy into a new data set.
就是这样。好了，我们坐下吧。非常容易。就像，我写下来。你明白这个原理，对吧？这就是你如何绘制参考。如何将数据从一个副本拉到一个新的数据集中。

02:20:30
The principles are the same.
原理是一样的。

02:20:31
As a book for.
作为一本书。

02:20:35
But opens it does that, yes, is that, like, can you find online, can you found how to cite this kind of?
但是打开它，是的，就像，你能在网上找到，你能找到如何引用这种吗?

02:20:45
Of source things. Let me tell you, let me tell you a story.
来源的东西。让我给你们讲个故事。

02:20:49
Okay, I'm sorry about what happened with this dataset. So, you see, I have several datasets, right?
好吧，对于数据集的问题我很抱歉。你看，我有几个数据集，对吧?

02:21:05
Absolutely. Let's set that are put from multiple places on the Internet.
绝对的。让我们把它们放在互联网上的多个地方。

02:21:12
Two weeks down the line, doctor gave message me, and she was like, you know.

两周后，医生给我留言，她说，你知道的。

02:21:18
I can't find the data anymore.
我再也找不到数据了。

02:21:21
What has gone off?
出了什么事?

02:21:23
I said, well, it's a good thing. We saved it right. We saved a copy and we saved the attribution statement.
我说，这是件好事。我们保存得很好。我们保存了一份副本，保存了归属声明。

02:21:31
So usually, when people make a data set, they will tell you how you ought to attribute the source of that. So I copied out the attribution statement. I copied out when we downloaded it.
通常，当人们做了一个数据集，他们会告诉你应该如何确定数据的来源。所以我把归属声明抄了出来。我下载的时候抄的。

02:21:41
I copied out the exact website, the name and everything, documentation, really important, right? Because you never know tomorrow their website goes down, the person decides not to do this anymore.
我把网站，名字和所有东西都抄了出来，文档，非常重要，对吧？因为你永远不知道明天他们的网站会崩溃，这个人决定不再这样做了。

02:21:55
It's gone.
这是一去不复返了。

02:21:56

Documentation. Keep a running text file like this. You see. Divided this data, this data set comes from this page.
文档。保持像这样的文本文件。你看到的。除以这些数据，这个数据集来自这个页面。

02:22:07
Very important trace of steps. Because one year on DR came back to this and asked, who is this from? There was one TXC file with all the information that you need.
非常重要的步骤。因为一年之后DR回来问，这是谁寄来的？有一个TXC文件包含了你需要的所有信息。

02:22:23
So how did you cite this? So I copied the attribution statement. So you use the attribution statement in your citation so you can So you look at Chicago, all the standards will tell you how to cite data sets, because people do.
你是怎么引用这个的？所以我复制了归因声明。所以你在引用中使用归因声明，所以你可以看看芝加哥，所有的标准都会告诉你如何引用数据集，因为人们会这么做。

02:22:40
So the specific way of setting data sets, but then you also need to acknowledge, based on the terms of use for specific data assets, sometimes is you need to attribute like This.
所以设置数据集的具体方式，但你也需要承认，基于特定数据资产的使用条款，有时你需要这样的属性。

02:22:50
Then you just copy and please, entire traditions and do technologies.
然后你只是复制和取悦，整个传统和技术。

02:22:55
It can be. This is why exercise date is important.
它可以是。这就是为什么锻炼日期很重要。

02:23:07
And this is why, when you do publish a thing, right? So for her, she didn't need to because the data wasn't important to replicate necessarily.
这就是为什么，当你发表一篇文章时，对吧？所以对她来说，她不需要这样做，因为数据对复制并不重要。

02:23:16
It's not the basis of an analysis. If she was using it to calculate the distances between this and this, then then the best practice would be, I need to upload that data set.
这不是分析的基础。如果她用它来计算这两点之间的距离，那么最好的做法是，我需要上传那个数据集。

02:23:27
Into a data repository, usually is an institutional repository, so that people can go back to it and then can double check your calculation.
进入一个数据存储库，通常是一个机构存储库，这样人们可以回到它，然后可以再次检查你的计算。

02:23:33
But she was just using it for visualization purpose.
但她只是为了可视化的目的。

02:23:37
Her research question was wartime China and economics, and so supply lengths during, I think, world War Two.
她的研究问题是战时的中国和经济，以及二战期间的供给长度。

02:23:47
So she's looking at like rice and greens, and how that travels along different supply pathways, railways and waterways.
所以她正在研究大米和蔬菜，以及它们是如何沿着不同的供应途径，铁路和水路运输的。

02:23:56
So the economic is a war history, but it's not like confrontation, just it's the other parts of war, the socioeconomic aspects of it.
所以经济是一部战争史，但它不像对抗，只是战争的其他部分，它的社会经济方面。

02:24:07
Which is a special question if you think about it, right? When, how far is something? Are your supplies from there, from your front line, right? People are thinking strategically about how they want to cut off supply lines.
如果你仔细想想，这是一个特别的问题，对吧？什么时候，有多远？你的补给是从前线来的，对吧？人们正在战略性地考虑如何切断供应线。

02:24:21
Proximity methods, right? The terrain methods, right? Something can be 15 came from you, but it's over mountain.
邻近方法，对吧？地形方法，对吧？有些事可能来自你，但它已经越过了山。

02:24:29
Right connection just accompanies where the rights are. That's where the soldiers are.
正确的联系只是伴随着权利的存在。那里有士兵。

02:24:41
So that's another This is what she did. I just want to remember from what people told me about. Okay, that's very good as a good proxy, right? If you have information about something and you can use it to.
这是另一个，这是她所做的。我只是想记住人们告诉我的事情。这是一个很好的代理，对吧？如果你有关于某事的信息，你可以用它来。

02:24:51
Make a educated So extension, extrapolation.
做一个有教养的推广，外推。

02:24:56
Yes, that's curious. Like to what degree to this in your experience when historians is gis, right? What degree do they actually use the analysis tools, which a lot of other patterns use? So things like so that statistics are coming important because, because I imagine, like, the data use is a bit different. So.
是的，这很奇怪。在你的经验中，当历史学家是gis时，这在多大程度上是这样的？他们实际上在多大程度上使用了分析工具，而许多其他模式都在使用这些工具？所以统计数据变得越来越重要，因为我想，数据的使用有点不同。所以。

02:25:13
Right? Yes, sometimes all we don't use it as often some people do. It depends on what questions you ask. So E.G., is it possible, how long would it take for someone to travel from Shanghai to.
对吧?是的，有时候我们不像有些人那样经常使用它。这取决于你问什么问题。例如，有没有可能，从上海到……

02:25:27
Chongqing, right? What's the easiest route? So what you could do is you could put in and say, reverse will take you this amount of time to travel per day, right? You can travel as much a day versus you take this route, this route, the cost of.
重庆,对吧?最简单的路线是什么？所以你可以做的是，你可以输入，反向每天会花费你这么多的时间，对吧？你可以一天走这么多路而不是走这条路，这条路的成本是。

02:25:42
Traveling is this much? They can calculate what's the most easiest way to get there, what's the shortest time you would have taken. You need to make a lot of assumptions like, it's a uniform travel time, right?
旅行就这么多吗？他们可以计算出到达那里最简单的方法，最短的时间是什么。你需要做很多假设，比如，这是一个统一的旅行时间，对吧？

02:25:52
You're not going to work with currents that run against you, so on and so forth, right? But you're making certain assumptions. So you can use some of these tools to help you. You're always talking about sidelines, right?
你不会和逆流而上的水流一起工作，等等，对吧？但是你做了一些假设。所以你可以使用这些工具来帮助你。你总在说场边的事，对吧？

02:26:03
So can you see something, right? So there are lots of projects where, E.G., there's one about a Greek island, and then they have powers like God powers. And then the question is, how far can these God towers see?
你能看到什么吗？所以有很多项目，比如，有一个关于希腊岛屿的，然后他们有像上帝的力量。接下来的问题是，这些上帝塔能看到多远？

02:26:15
Right? So you can use the elevation and you can sort of use what we call viewshed, right? Like a watershed of viewshed. What can you see, right?
对吧?你可以使用仰角也可以使用我们所说的视域，对吧？像一道分水岭。你能看到什么，对吧？

02:26:23
So is it possible that someone would have seen the ship go by if they were on this gut tower? That's a historical question, which you can calculate, re enact, right? You know, the height of the tower.
如果有人在这个塔上有没有可能看到船经过？这是一个历史问题，你可以计算，重新制定，对吧？你知道的，就是塔的高度。

02:26:35
You know exactly where it was, you know the weather conditions, or you assume what the weather conditions are. And then you can, you can tell.
你知道它在哪里，你知道天气状况，或者你假设天气状况是什么。然后你就知道了。

02:26:44
Reasonably, whether something can be seen.

合理地说，是否可以看到一些东西。

02:26:46
I guess. But the Titan, very interestingly, because you've got all these ships, and then they're redoing positions and stuff, so you could actually reconstruct geographical.
我猜。但是泰坦，非常有趣，因为你有所有这些船，然后他们重新定位和东西，所以你实际上可以重建地理。

02:27:00
Proximities of all the different responders and the ship itself, and potentially that where the ice work was. We actually do know the information.
所有不同的救援人员和船本身的距离，可能还有冰的位置。我们确实知道这些信息。

02:27:09
First reason someone's done the analysis, right? I've had previous students work on projects like a PW ship that was going up towards Japan during World War II.
有人做分析的第一个原因，对吧？我以前有学生做过一个项目，比如二战期间开往日本的战舰。

02:27:20
Sung by an American submarine right off the coast of China. And then you can actually track where the PWS ended up the escape, and they ended up on mainland China, right? And then the Japanese officers were picked up by a Japanese Navy ship.
一艘美国潜艇在中国海岸附近演唱。然后你就可以追踪到PWS最终逃到哪里去了，他们最终到了中国大陆，对吧？然后日本军官被一艘日本海军军舰接走。

02:27:32
You went to Tokyo. So you can use schoolly telling special data to tell that story, to emplace it and plot it geographically.

你去了东京。所以你可以用学校里的特殊数据来讲述这个故事，把它放在地理位置上。

02:27:42
So all the oral histories, or the accounts, or the newspaper reports, right? Makes sense geography, because now you're cooler keeping them in the same space.
所以所有的口述历史，或账目，或报纸报道，对吧？这在地理上说得通，因为现在把它们放在同一个地方会更酷。

02:27:49
Yes, just act as like secondary evidence to make sure that all history is correct. It depends on your question, right? Maybe your hypothesis is this person is flying.
是的，就像次要证据一样，确保所有的历史都是正确的。这取决于你的问题，对吧？也许你的假设是这个人在飞。

02:28:04
Right? This person says he managed to swim from the spot of the sinking to the coast, right? Okay, so let's figure that out. Where did it sink?
对吧?这个人说他设法从沉船地点游到了海岸，对吗？好的，我们来算一下。它在哪里沉没了？

02:28:13
How far was it from the coast? Is it reasonable to say that somebody could spend that distance in that amount of time without dying. So then that's your primary source. Your special analysis becomes your primary tool of analysis.
它离海岸有多远？有人能在这么长的时间内度过这么长的距离而不死亡，这合理吗？这就是你的主要来源。你的特殊分析成为你分析的主要工具。

02:28:25
Right. So the distinction room primary and secondary depends on your question.
正确的。所以区分主要房间和次要房间取决于你的问题。

02:28:32
Okay.
好吧。

02:28:38
Was that fun? You see why? I like special data.
好玩吗？明白为什么了吗？我喜欢特殊的数据。

02:28:43
Okay.
好吧。

02:28:45
So what we've done, actually is we've done two things that historians tend to do with jazz. One is we try and recover information from maps, historical maps, right? That wouldn't otherwise exist in.
所以我们所做的，实际上是我们做了两件历史学家倾向于对爵士乐做的事。一种是我们尝试从地图，历史地图中恢复信息，对吧？这是不存在的。

02:29:02
Textual form, E.G., right? So this special data that's locked away in paper sources, we can recover them, right? And we can just oppose them against other things.
文本形式，例如，对吧？所以这些被锁在纸上的特殊数据，我们可以恢复它们，对吧？我们可以把它们和其他东西对立起来。

02:29:11
The other thing that we do that we do is with this, E.G., is to create visualizations that tell a story, or that help communicate some geographical reality.
我们做的另一件事是用这个，例如，创建可视化来讲述一个故事，或者帮助传达一些地理现实。

02:29:22

The reality that you want to communicate, right? In this case, doctor was trying to show sort of the proximal distance between the different cities, and how they connect the federal water Risk.
你想要交流的现实，对吧？在这个案例中，博士试图展示不同城市之间的近距离，以及它们是如何连接联邦水风险的。

02:29:31
So that's the sorry she wanted to tell. And so historians can use your ass to create these specializations.
这就是她想说的抱歉。所以历史学家可以利用你的屁股来创造这些专业。

02:29:36
Okay.
好吧。

02:29:39
Right? So when there, but this is what we've done today. We've looked at.
对吧?这就是我们今天所做的。我们已经看过。

02:29:48
So you can geographical history.
所以你可以学习地理历史。

02:29:50
What it means for count the questions that we do, right? Um, we look at GS, historical. GS. Okay. And.
计算我们做的问题意味着什么，对吧？我们看看GS的历史。GS。好吧。和。

02:30:00
The limitations, right? That come with the technology itself, which are substantial, but also there are ways to sort of recover.
限制，对吧？这是技术本身带来的，这是实质性的，但也有一些方法可以恢复。

02:30:09

Against that.
反对这一点。

02:30:11
And so we did come to two things that historians tend to do with GIS, which is to capture data right your reference, to get the data recovered from the labs.
因此，我们得出了历史学家倾向于用GIS做的两件事，一是获取你所参考的数据，二是从实验室恢复的数据。

02:30:23
Or to produce snacks, which are amazing sheets, right?
或者制作零食，这些都是很棒的床单，对吧?

02:30:27
Obviously, there's a lot more, so you can explore. So please do next week.
显然，还有很多，所以你可以探索。所以请下周再做。

02:30:32
What we're going to do is the presentations, just to check in in your project. Okay.
我们要做的是演示，这是一个检查。我是说，去吧